

CoCaTS: A Cooperative Caching-Enabled Transmission Scheme in Ultra-Dense LEO Satellite Networks

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Abstract—In recent years, ultra-dense Low Earth Orbit (LEO) satellite networks such as Starlink have been rapidly developed, unparalleled advantages in terms of seamless coverage, multi-gigabit bandwidth, and low latency make it attractive to 6G system to provide global seamless Internet access service. However, inherent challenges such as on-board resource limitation and backhaul connectivity constraint render LEO satellite networks inadequate to satisfy the requirements of resource-consuming applications. To tackle these challenges, in this paper, we devise a Cooperative Caching-enabled Transmission Scheme (CoCaTS). Considering the constraints of storage space, power consumption and user-satellite association, we formulate a joint optimization problem and adopt block coordinate descent approach to decouple the original NP-hard problem as solvable subproblems. In specific, we propose a greedy-based algorithm to obtain the optimal caching placement, a simple allocation algorithm that minimizes the transmission power for users with theoretical proof, and a distributed many-to-many matching algorithm to obtain Nash equilibrium solution for user-satellite association. Extensive evaluations are conducted to demonstrate the superiority of CoCaTS, and results show that, compared to the representative baselines, CoCaTS can achieve higher traffic offloading, lower latency, and lower cost in terms of transmission power and on-board storage.

Index Terms—Cooperative transmission, coded caching, ultra-dense LEO satellite networks.

I. INTRODUCTION

DUE to the unparalleled advantages in terms of ubiquitous coverage, multi-gigabit bandwidth, and low latency [1],

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[2], Low Earth Orbit (LEO) satellite networks can provide ubiquitous Internet access service without the requirement of terrestrial infrastructure, which becomes attractive for global users especially in rural areas and transoceanic communications [2], [3], [4]. Nowadays, pioneering companies including SpaceX [5] and OneWeb [6] are devoted to constructing mega constellation with over 10,000 LEO satellites and achieve stage-wise progress [3], [7]. For example, Starlink system, led by SpaceX, has launched over 4,000 LEO satellites in total and provides high-speed Internet access and cellular services to millions of subscribers [5], [8].

Although the scale and ability of LEO satellite networks have been significantly developed compared to existing ones in early stage like Iridium system, the performance bottleneck still exists facing resource-consuming applications, e.g., high-bandwidth multimedia streaming, latency-sensitive content distribution [2], [9], [10]. The bottleneck issue is mainly constrained by two fundamental limitations: On the one hand, inherent resource-limited environment of LEO satellites significantly hinders the network capacity, for example, LEO satellites in Starlink system often encounter restrictions including limited return link bandwidth, energy consumption issues, and intense competition among terrestrial users, leading to low overall system efficiency and a capped number of accessible users [2], [10], [11]. On the other hand, due to the geographical restriction for building ground station infrastructure, the backhaul connectivity between LEO satellites and Ground Stations (GSs) cannot be globally guaranteed. The infrastructure limitations makes inter-satellite communications become necessary to complete end-to-end transmission from terrestrial users to server, leading to additional resource overhead and latency. These practical constraints render current LEO satellite networks inadequate for meeting the growing requirements in terms of bandwidth and latency from users' resource-consuming applications.

To alleviate the bottleneck issue of LEO satellite networks, one promising solution is leveraging storage resource to conserve scarce communication resource. In LEO satellite networks, caching-based solutions enable direct content delivery from satellite nodes, avoiding traditional bent-pipe transmission through backhaul connection [12], [13], as a result, significant performance improvement can be achieved through on-board caching, which provides substantial benefit to the resource-limited environment of LEO satellites. Pioneering explorations

consider basic caching models and validate the effectiveness of on-board caching [14]. Following that, existing studies put numerous efforts through caching placement [15], [16], joint resource optimization [17], [18], [19], Quality-of-Service (QoS) optimization [13], [20]. To further address inherent limitations of single-node caching in dynamic orbital environments, recent studies further extend traditional caching scheme to practical one through cooperative caching and transmission [2], [16], and results demonstrate that such design can acquire high-quality content delivery service with low latency and bandwidth consumption.

Unfortunately, existing caching-based solutions nevertheless confront two critical challenges, leading to performance bottleneck along with exponential content explosion:

1) *Content Correlation Constraint*. Traditional caching schemes demand content integrity, which means each satellite node has to reduplicative cache an entire content file to acquire caching gain to users, thus failing to fully exploit heterogeneous node caching capabilities [2], [16], [21]. This restriction caps the upper bound of performance gains from caching, while simultaneously causing redundant storage and substantial routing lookup cost.

2) *Spatial Association Constraint*. Traditional transmission scheme only considers single association between the target user and LEO satellite, once the satellite node with desired content is far away from the target user, the transmission cost might significantly increase or even unreachable, which significantly hinders the spatial benefit from ultra-dense LEO satellites. Although some existing studies consider cooperative transmission in satellite networks, these solutions also bring additional communication overhead such as spectrum and interference challenges in open-world scenario. Thus, to push the bottleneck limit of LEO satellite networks facing resource-consuming applications, it is necessary to design a novel transmission scheme that can address these challenges.

To tackle the aforementioned two critical challenges, in this paper, we propose a Cooperative Caching-enabled Transmission Scheme (CoCaTS) for ultra-dense LEO satellite networks, which fully leverage the storage resource to achieve efficient traffic offloading. The basic idea behind the proposed scheme can be described as follows. First, CoCaTS adopts coded caching technique to eliminate the *Content Correlation Constraint* and thus enables fine-grained content block. Second, CoCaTS exploits the spatial diversity of multi-point cooperation transmission scheme by using Non-Orthogonal Multiple Access (NOMA) technique to eliminate *Spatial Association Constraint*. Specifically, we formulate a joint optimization problem to maximize the performance of traffic offloading involving three variables, i.e., caching placement, power allocation, and user-satellite association. Considering the difficulty of solving the joint optimization problem, i.e., Mixed-Integer Non-Linear Programming (MINLP) problem, we further adopt Block Coordinate Descent (BCD) method to decouple variables and efficiently obtain the solution for each subproblem. For caching placement, we propose a greedy-based algorithm to obtain the optimal solution. For the power allocation, we propose a simple allocation algorithm that minimizes the transmission power for

users with theoretical proof. For user-satellite association, we propose a distributed many-to-many matching algorithm to obtain stable solution that converges to a Nash equilibrium. Extensive evaluations are conducted to demonstrate the superiority of the proposed CoCaTS. Results show that CoCaTS significantly reduces both bandwidth consumption and file download time compared to representative baselines, and exhibits outstanding convergence in dynamic scenarios, which provides a better QoS for resource-consuming applications in ultra-dense LEO satellite networks.

The main contributions of this paper are summarized as follows:

- To tackle the performance bottleneck problem facing resource-consuming applications, we devise a novel transmission scheme, called CoCaTS, for ultra-dense LEO satellite networks. CoCaTS adopts coded caching and NOMA techniques to fully exploit satellite storage capacity and multi-access capability, and thus break through both content correlation constraint and spatial constraint existing in traditional schemes.
- To tackle the NP-hard optimization problem, we adopt BCD method to decouple variables and devise efficient algorithms respectively. For caching placement, we propose a greedy-based algorithm to obtain the optimal solution. For power allocation, we propose an efficient allocation algorithm that minimizes the transmission power with theoretical proof. For user-satellite association, we propose a distributed many-to-many matching algorithm to obtain stable solution that converges to Nash equilibrium.
- We conduct extensive evaluations in various scenarios based on realistic network settings. Results show that CoCaTS not only significantly outperforms traditional schemes in terms of offloading traffic, file download time, and resource cost, but also provides near-optimal performance with convergence in dynamic scenarios.

The rest of this paper is organized as follows. At first, the related studies regarding caching-based solutions in satellite networks are discussed in Section II. After that, the system model and problem formulation are given in Section III. To efficiently solve the proposed MINLP optimization problem, the caching placement problem is discussed in Section IV, and user-satellite association and power allocation are discussed in Section V. At last, the performance evaluation is conducted in Section VI and conclusions are drawn in Section VII.

Symbol Notation: Boldface lowercase and uppercase letters denote vectors and matrices, respectively. $\mathbf{a}^T$ denotes transposition of vector $\mathbf{a}$. For a set $\mathcal{S}$, $|\mathcal{S}|$ denotes the size of elements in the set. The set of n -by- m real matrices and natural matrices are denoted as $\mathbb{R}^{n \times m}$ and $\mathbb{N}^{n \times m}$, respectively. $\mathcal{N}(\mu, \Sigma)$ denotes the Gaussian distribution with mean μ and covariance matrix Σ .

II. RELATED WORK AND MOTIVATION

In this section, we first introduce the existing studies in recent years, and then discuss the motivation of this work based on an intuitive performance comparison with illustration.

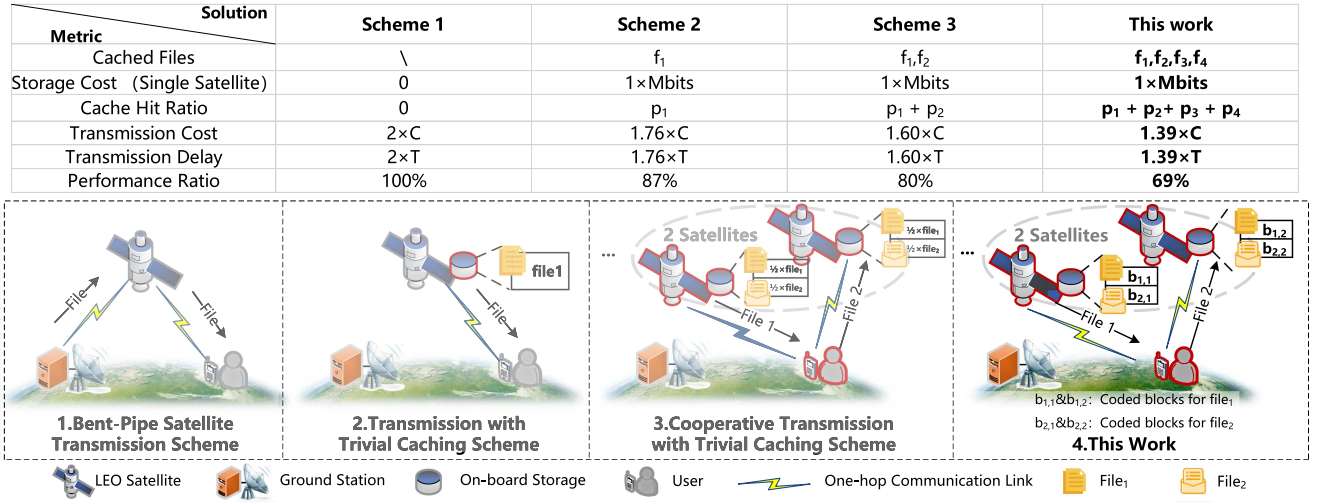


Fig. 1. Performance comparison of four typical transmission schemes in LEO satellite networks, including bent-pipe satellite transmission, transmission with trivial caching, cooperative transmission with trivial caching, and cooperative caching-enabled transmission (coded caching scheme adopted).

A. Related Work

The approach of leveraging satellite storage resources to conserve scarce communication resources and enhance users' QoS mainly involves two aspects of research: satellite caching strategies and user access optimization. In the following, we introduce existing studies from these two aspects.

As the scale of satellite deployments grows and on-board storage resources increase, it becomes increasingly feasible to improve users' QoS using in-network caching technologies in LEO satellite networks [12]. A large number of studies indicate that using in-network caching can enhance the performance of LEO satellite networks and improve users' QoS [22]. Urs et al. [23] were among the first to propose caching strategies in wireless networks, improving content delivery quality in high-loss networks. Inspired by this, Wu et al. [14] further introduced caching techniques into satellite networks and jointly optimized caching decisions for both satellites and GSs, reducing satellite bandwidth consumption. Since then, a variety of caching strategies tailored to LEO satellite networks have emerged. Jiang et al. [13] proposed a density-based network division algorithm and a cache updating mechanism under a time-varying network model. However, the area partitioning method in this scheme fails to capture user request characteristics, which is unfavorable for satellite caching decisions. Therefore, Tang et al. [2] developed a cooperative caching scheme that incorporates a regional feature prediction model to update user preference information, thereby partitioning user regions. Mao et al. [21] further leveraged advanced deep reinforcement learning techniques to design a hierarchical caching scheme. All of the above works suffer from content correlation issues. They must cache an entire content, which leads to wasted storage resources and requires efficient content lookup mechanisms, resulting in additional routing cost and lookup conflicts.

In addition to improved caching strategies, optimizing user access strategies can lead to more efficient utilization of satellite communication and caching resources, thereby further

enhancing the users' QoS. Therefore, many researchers proposed various solutions in satellite networks. In terms of content lookup, Tang et al. [24] proposed an opportunistic content-aware routing scheme combined with satellite caching. In terms of content offloading, Cao et al. [25] utilized edge computing, combining communication and computing resources, to formulate a joint optimization problem for offloading. In addition, Chen et al. [26] leveraged unmanned aerial vehicles to assist satellites in offloading and proposed a joint task offloading algorithm. In terms of failure handling, Jia et al. [27] addressed the problem of service recovery under resource failures in satellite networks. Single-link content offloading suffers from performance bottlenecks, as the target content may reside in distant nodes, leading to higher communication cost and longer download time. Therefore, Zhang et al. [28] studied multi-satellite cooperative transmission based on orthogonal time-frequency space. The aforementioned user access studies are limited by content correlation, making it difficult to realize multi-satellite cooperative offloading and thus wasting the advantages of ultra-dense satellite deployment. On the other hand, multi-satellite cooperative transmission does not involve in-network caching and falls under traditional bent-pipe transmission, which incurs high resource costs and latency.

B. Motivation

To motivate this work and discuss the shortcoming of the traditional solutions, as shown in Fig. 1, we theoretically illustrate the performance comparison of four typical transmission schemes in LEO satellite networks, i.e., bent-pipe transmission scheme, transmission scheme with trivial caching scheme, cooperative transmission with trivial caching scheme, and the one considered in this work. To simplify complicated details in each scheme and intuitively present a rough performance result, we assume that the one-hop communication cost for an entire file is C and the transmission delay is T . Meanwhile, the total content library $\mathcal{F}$

contains 10 files, which follows Zipf distribution with shape parameter 0.69 from UCB traces provided in [29], and storage cost is M bits if an entire file is cached on-board. For caching-enabled schemes, i.e., scheme 2-4, p_i denotes cache hit ratio for i -th file, which can be calculated by Zipf distribution in (1). In specific, the corresponding cache hit ratio follows $\{24.5\%, 15\%, 11.7\%, 9.7\%, 8.3\%, 7.7\%, 6.5\%, 6\%, 5.5\%, 5.1\%\}$ from the first file to 10-th file. If multiple files can be accessed from on-board storage, then the cumulative cache hit ratio can be calculated by $\sum_{i \in \mathcal{F}^*} p_i$, $\mathcal{F}^* \subseteq \mathcal{F}$ for cached file set $\mathcal{F}^*$.

As shown in Fig. 1, we consider the expected performance for one file access attempt from a typical user. First, for the baseline (i.e., scheme 1), each attempt experiences 2-hop transmission without on-board storage, and the expected transmission cost is $2 \times C$. Second, for trivial caching scheme (i.e., scheme 2), the cost can be calculated by $(1 - \sum_{i \in \mathcal{F}^*} p_i) \times 2 \times C$, and the corresponding cost significantly reduces to $1.76 \times C$ with 13% degradation by only caching a single file. However, such caching gain is also the upper bound of scheme 2, the *Spatial Association Constraint* hinders further its performance improvement even under an ultra-dense LEO satellite deployment environment. Third, for cooperative transmission with trivial caching scheme (i.e., scheme 3), this cost can further reduce to $1.60 \times C$ since two times storage space can be leveraged thanks to multiple satellite association. However, due to dynamic nature of LEO satellite movement, this scheme can hardly resolve *Content Correlation Constraint* and also introduces additional communication resource occupation such as spectrum for terrestrial-satellite transmission. Instead, for cooperative caching-enabled transmission scheme (i.e., scheme 4), we consider two solutions to address the aforementioned constraints. The first one is coded caching based design, which eliminates the *Content Correlation Constraint* and allows various satellites to cache fine-grained content block. The second one is Non-Orthogonal Multiple Access (NOMA) based communication design, which eliminate *Spatial Association Constraint* and allows a single user access multiple LEO satellite without using extra spectrum resource. In this case, scheme 4 can reach the minimum expected cost, i.e., $1.39 \times C$, and transmission delay, i.e., $1.39 \times T$, but consumes the same storage space without additional resource occupation such as spectrum. Motivated by the observation as shown in Fig. 1, in this paper, we propose CoCaTS to address the inherent limitations of existing studies to reduce transmission cost such as power and enhance the transmission performance in ultra-dense LEO satellite networks.

III. PROBLEM FORMULATION

In this section, we first introduce the system model considered in this work, including transmission procedure, caching model, and channel model, respectively. After that, we formulate a joint optimization problem that considers three critical variables, i.e., caching placement, power allocation, and user-satellite association. Finally, we present an overview of CoCaTS. The main symbol used in this paper and their corresponding definitions are listed in Table I.

TABLE I
SYMBOL DEFINITION

Symbol	Definition
$\mathcal{S}$	The set of LEO satellites in the network.
$\mathcal{U}$	The set of terrestrial users served by the satellite network.
$\mathcal{F}$	The set of content library.
$\mathbf{P}$	The satellite transmitter output power allocation matrix.
$p_{u,s,i}$	The allocated transmitter output power for the s -th satellite serving user u with the i -th file.
$\mathbf{A}$	The user-satellite association matrix.
$a_{u,s}$	The user-satellite association between the s -th satellite and user u .
$\mathbf{m}$	The caching placement vector for all LEO satellites.
m_i	The caching placement for the i -th file.
M	The on-board storage space on each LEO satellite.
K_u	The number of LEO satellites that provides service for user u .
$\tilde{p}_i$	The popularity of the i -th file.
O_u	The offloaded traffic of user u .
Φ	The matching pairs of users and satellites.
T	The update period for CoCaTS to reconfigure onboard decisions including caching placement, power allocation, and user-satellite association.
d	The frame duration for user-satellite communication.

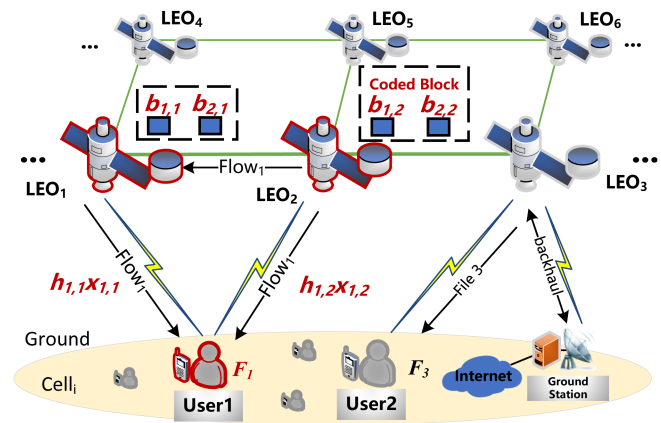


Fig. 2. Illustration of NOMA based cooperative transmission mechanism in ultra-dense LEO satellite networks.

A. System Model

In this paper, we consider a downlink cooperative transmission mechanism in ultra-dense LEO satellite network. As illustrated in Fig. 2, terrestrial users are covered by multiple LEO satellites simultaneously, each LEO satellite is equipped with an on-board storage unit. To access LEO satellite networks and establish the communication link, the terrestrial user can search for periodic broadcast signal from LEO satellites and send

the request to setup a connection [13], [30]. Once the identity authentication is accomplished, the terrestrial user can access Internet and request interested content through satellite-terrestrial link. Without loss of generality, we focus on the analysis of one certain geographical area that $K^{total} = |\mathcal{S}|$ LEO satellites' coverage overlap on the ground, and $\mathcal{S}$ and $\mathcal{U}$ are used to respectively denote all satellites that provide coverage in the area and all users located inside the area.

The specific transmission procedure for serving resource-consuming applications in LEO satellite networks can be described as follows. For a terrestrial user u , he/she first sends a request for the i -th file to an associated LEO satellite, and the satellite coordinates with available satellites to transmit their cached content of the i -th file to user u if i -th file is cached on-board. Otherwise, the satellite only forwards the request to the GS to retrieve the requested file from the destined server on the Internet. Note that multiple satellites serve the same user for the i -th file simultaneously over overlapping frequency band via NOMA [10], [31]. If the i -th file can be recovered from received signals at user u , the request is satisfied; Otherwise, the rest of the requested content file are retrieved from the GS through satellite-GS backhaul link. To intuitively illustrate the procedure, as shown in Fig. 2, the terrestrial User 1 requests file F_1 from two LEO satellites, LEO_1 and LEO_2 are selected to serve the user with cached data (i.e., coded blocks $b_{1,1}$ - $b_{1,2}$). In specific, since the wireless signals $x_{1,1}$ and $x_{1,2}$ are respectively transmitted through the same spectrum resource experiencing channel condition $h_{1,1}$ and $h_{1,2}$, the user decodes the signal from LEO_1 and LEO_2 by utilizing Successive Interference Cancellation (SIC) receiver, and reconstruct file F_1 after decoding. In contrast, User 2 requests file F_3 . However, since the satellite does not cache F_3 , it has to retrieve the complete file content from the ground station via the backhaul link.

1) Caching Model: For various resource-consuming applications, without loss of generality, we consider a content library denoted by $\mathcal{F}$ consisting of $|\mathcal{F}|$ files. The file popularity is modeled as the Zipf distribution with shape parameter θ , and then the popularity of the i -th ranked file can be calculated by

$$\tilde{p}_i = \frac{1/i^\theta}{\sum_{n=1}^N 1/n^\theta}. \quad (1)$$

In the same geographical area, we consider that each user requests for the i -th file with the same probability of popularity distribution, i.e., $\tilde{p}_i = \tilde{p}_{u,i}$, which denotes the request probability of each user for the i -th file.

In order to address the *Content Correlation Constraint*, a coded caching scheme similar to [18] is adopted. The on-board storage space of each satellite is considered as the same one with M bits. All files are assumed to have the same size and each one is split into $c \in \mathbb{N}^+$ equal-sized original blocks, where each one is with the size of $L = L^o + L^c$, L^o denotes the size of an original block and L^c denotes the overhead of coding. After that, c original blocks of each file are encoded into an arbitrarily large number of coded blocks to be cached by satellites. By utilizing coded caching, when a user receives an arbitrary number n ($n \geq c$) of blocks from satellites, he/she can recover the original

file from the received blocks. By carefully selecting coding coefficients, each satellite can provide linearly independent blocks to the user with high probability. In practice, it can be achieved through Maximum-Distance Separable (MDS) [18] and Random Linear Network Coding (RLNC) codes [10].

2) Channel Model: A NOMA based cooperative transmission scheme similar to [10] is adopted to address the *Spatial Association Constraint*. We consider the downlink channel of satellite-terrestrial link, and well-accepted Shadowed Rice fading model is adopted [32, eq. (4)]. Each LEO satellite is equipped with N phased array beamforming antennas [33], and beamforming vector is denoted by $\mathbf{w} = [w_1, w_2, \dots, w_N]^T$, which follows $|w_n| = 1$. The channel gain of satellite-terrestrial link between user u and the s -th satellite is considered as $\mathbf{h}_{u,s} \in \mathbb{C}^{1 \times N}$. Each terrestrial user communicates with LEO satellites through a time-synchronized manner and length of frame duration is denoted by d . The channel state of satellite-terrestrial link at each frame duration is invariant but randomly changes from one frame duration to another [31]. Users can obtain the instantaneous exact Channel State Information (CSI) as the basis for theoretical analysis. For user u , there are K_u LEO satellites to provide access service simultaneously, and then the signal transmitted by the s -th satellite can be presented by

$$\mathbf{x}_{u,s,i} = \sqrt{p_{u,s,i}} \mathbf{w}_{u,s} y_i, \quad (2)$$

where y_i denotes the symbol transmitted from the satellite for the i -th file, $p_{u,s,i}$ denotes the transmitter output power for the s -th satellite serving user u with the i -th file, $\mathbf{w}_{u,s} \in \mathbb{C}^{1 \times N}$ denotes the beamforming vector for user u transmitted by the s -th satellite. The received signal at the user is given by

$$\hat{y}_{u,s,i} = \sqrt{p_{u,k,i}} \mathbf{h}_{u,s} \mathbf{w}_{u,s} y_i + n_0, \quad (3)$$

where $n_0 \sim \mathcal{CN}(0, N_0)$ is the complex Additive White Gaussian Noise (AWGN) of power N_0 . The Signal-to-Interference-plus-Noise-Ratio (SINR) of received signal from the s -th satellite at user u can be calculated by

$$SINR_{u,s} = \frac{p_{u,s,i} |\mathbf{h}_{u,s} \mathbf{w}_{u,s}|^2}{\sum_{j \neq s} p_{u,j,i} |\mathbf{h}_{u,j} \mathbf{w}_{u,j}|^2 + N_0}. \quad (4)$$

Based on Shannon equation, we can have transmission power threshold for successfully decoding the signal from the s -th satellite by substituting (4) into Shannon equation $R_{u,s} = B \log(1 + SINR_{u,s})$ as

$$\frac{p_{u,s,i} |\mathbf{h}_{u,s} \mathbf{w}_{u,s}|^2}{\sum_{j \neq s} p_{u,j,i} |\mathbf{h}_{u,j} \mathbf{w}_{u,j}|^2 + N_0} \geq \gamma_i, \quad (5)$$

$$p_{u,s,i} = \frac{\gamma_i (\sum_{j \neq s} p_{u,j,i} |\mathbf{h}_{u,j} \mathbf{w}_{u,j}|^2 + N_0)}{|\mathbf{h}_{u,s} \mathbf{w}_{u,s}|^2}, \quad (6)$$

where $\gamma_i = 2^{(\frac{m_i L}{Bd})} - 1$ for notation simplicity, and B denotes bandwidth of satellite-terrestrial link.

In a practical LEO satellite system, inter-satellite interference management is actually a critical concern, especially for dense constellations. Existing studies usually consider two solutions to mitigate inter-satellite interference in a multibeam antenna-equipped LEO satellite system [34, Sec. V], i.e., dynamic handover, and joint beam scheduling and power allocation. The

former adopts an antenna tracking strategy to dynamically handover access satellites to mitigate interference. The latter often formulates a joint beam-power scheduling problem for overlapping coverage and dynamically determines beam hopping, frequency band selection, and transmitting power allocation. However, such a joint optimization problem is sophisticated. In this case, a simple but effective beam switching strategy such as [35] is preferred in this paper.

The proposed CoCaTS utilizes multi-connectivity characteristic in ultra-dense LEO satellites. While terrestrial users request content files from resource-consuming applications, they can receive superposition signals from multiple LEO satellites through NOMA technique, and then the traffic offloaded from satellite-GS backhaul link can be calculated by the number of successfully decoding signals through SIC receiver. In this context, we consider successful decoding situations while user u requests the i -th file and then the offloaded traffic of user u can be calculated by

$$O_u(\mathbf{A}, \mathbf{P}, \mathbf{m}) = \begin{cases} K_u m_i, \forall K_u < \left\lceil \frac{c}{m_i} \right\rceil, \\ L^o, \forall K_u \geq \left\lceil \frac{c}{m_i} \right\rceil, \end{cases} \quad (7)$$

where $K_u = \sum_{s \in \mathcal{S}} a_{u,s}$ is the number of satellites to which the user u is connected. $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_U]^T$ is the user-satellite association matrix and $\mathbf{a}_u = [a_{u,1}, a_{u,2}, \dots, a_{u,|\mathcal{S}|}]$ denotes the user-satellite association vector for user u , $\mathbf{P}$ is actually a three-dimensional matrix, for i -th file, $\mathbf{P}_i = [\mathbf{p}_{1,i}, \mathbf{p}_{2,i}, \dots, \mathbf{p}_{U,i}]^T$ is the satellite power allocation matrix and $\mathbf{p}_u = [p_{u,1,i}, p_{u,2,i}, \dots, p_{u,|\mathcal{S}|,i}]$ is the vector of satellite power allocation for user u , $\mathbf{m} = [m_1, m_2, \dots, m_{|\mathcal{F}|}]$ is the caching placement vector for all LEO satellites.

B. Problem Formulation

As illustrated in Fig. 1, caching-enabled transmission solutions can significantly acquire both cost degradation and performance improvement by leveraging the on-board storage, which is attracting for resource-limited environment especially LEO satellite networks. To further utilize the ultra-dense LEO satellites in ongoing mega constellation such as Starlink, in this paper, a cooperative caching-enabled transmission scheme called CoCaTS is considered, which can diminish the transmission cost regarding traffic consumption while improving transmission performance in terms of download delay. In specific, based on the aforementioned system model, we jointly consider three kinds of variables, i.e., caching placement, power allocation, and user-satellite association, respectively. In this case, we can formulate a joint optimization problem for maximizing offloading traffic:

$$\max_{\mathbf{A}, \mathbf{P}, \mathbf{m}} \sum_{i \in \mathcal{F}} \sum_{u \in \mathcal{U}} \tilde{p}_i O_u(\mathbf{A}, \mathbf{P}, \mathbf{m}) \quad (\text{P1})$$

$$s.t. \quad \|\mathbf{w}_{u,s}\| = 1, \forall u \in \mathcal{U}, \forall s \in \mathcal{S}, \quad (\text{C1-1})$$

$$\sum_{s \in \mathcal{S}_u} SINR_{u,s} \geq \gamma_i, \forall a_{u,s} = 1, \quad (\text{C1-2})$$

$$\sum_i m_i \leq M, \forall i \in \mathcal{F}, \quad (\text{C1-3})$$

Algorithm 1: CoCaTS Procedure.

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1 for each update period  $T$  do
2   # 1.Partition Area Update:
3   System updates partition results;
4    $\mathcal{S} \leftarrow$  Collect available satellites after partition area
    update;
5    $\mathcal{U}^{au} \leftarrow$  Collect active users for partition area;
6   # 2.Update caching and matching decisions:
7   Select a manager satellite for each partition area;
8   for each partition area do
9      $\mathbf{m} \leftarrow$  The manager executes Algorithm 2;
10     $\mathbf{A} \leftarrow$  The manager executes Algorithm 4;
11    The manager distributes these results to other
      satellites;
12    # 3.Real-time Power Allocation:
13    for each frame duration  $d$  do
14       $\mathcal{U} \leftarrow$  Receive requests from active users
         $\mathcal{U}^{au}$ ;
15      for each user  $u \in \mathcal{U}$  do
16         $\mathbf{P} \leftarrow$  Algorithm 3 ( $\mathcal{S}, \mathbf{m}$ );
17      end
18    end
19  end
20 end

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$$0 \leq m_i \leq c, \forall i \in \mathcal{F}, \quad (\text{C1-4})$$

$$P^{BB} + P^{RF} \leq P^{max}, \forall u \in \mathcal{U}, s \in \mathcal{S}, i \in \mathcal{F}, \quad (\text{C1-5})$$

$$K_u \leq \left\lceil \frac{c}{m_i} \right\rceil \leq K^{total}, \forall u \in \mathcal{U}, i \in \mathcal{F}, \quad (\text{C1-6})$$

$$a_{u,s} \in \{0, 1\}, p_{u,s,i} \in \mathbb{R}, m_i \in \mathbb{N}. \quad (\text{C1-7})$$

Specifically, constraint (C1-1) represents the beamforming constraint while serving each terrestrial user. Constraint (C1-2) indicates the received SINR at u -th user must satisfy the successful decoding threshold if user u and s -th LEO satellite are associated. Constraint (C1-3) implies that the total number of cached blocks cannot exceed the capacity on each LEO satellite. Constraint (C1-4) represents that the caching placement variable for each file cannot exceed the number of original blocks. Constraint (C1-5) limits the total power consumption to P^{max} for serving a single file. For a satellite digital communication system, the source signal requires onboard processing process through the baseband and RF module, where P^{BB} and $P^{RF} = \mathbf{a}_u \mathbf{P}_u = \sum_{s=1}^{|\mathcal{S}|} a_{u,s} p_{u,s,i}$ represent power consumption of the baseband module and RF module, respectively. Constraint (C1-6) represents the number of LEO satellites. Constraint (C1-7) limits the value range of variables, i.e., the association variable $a_{u,s}$ must be zero or one, the value of allocated power $p_{u,s,i}$ must be a real value, and the value of caching placement m_i must be a positive integer.

Problem **P1** is a Mixed-Integer Non-Linear Programming (MINLP) problem, and thus we can hardly derive the solution directly through commonly used optimization method such as

convex optimization. However, inspired by the alternating optimization approaches [10], [36], if we decouple the variables $\mathbf{m}$ and $\mathbf{a}$, $\mathbf{P}$, the original problem becomes solvable. Therefore, we adopt Block Coordinate Descent (BCD) method to decouple variables and efficiently obtain the optimal solution for each subproblem. To optimize a multi-variable objective in BCD method, we optimize the objective in terms of one of the coordinate blocks while the other blocks are fixed at each iteration. Thus, we decompose problem **P1** into two subproblems, i.e., caching placement subproblem and user-satellite association subproblem, and design Algorithm 1 to solve these subproblems iteratively. In specific, these subproblems are solved iteratively as described in Algorithms 2 and 4. It should be noted that two coupled variables exist in user-satellite association subproblem.

The classical convergence conditions of BCD require that the overall objective function is continuously differentiable with bounded level sets, the constraint set is a Cartesian product of closed convex sets (BCD-C1), and each subproblem has a unique solution under these conditions [37, Sec. 2.7]. Recent theoretical developments in nonconvex BCD methods relax these requirements, allowing convergence to a stable solution under finite sets, locally bounded nonconvex or non-differentiable conditions (BCD-C2) [38], [39], [40]. The objective function in problem **P1** satisfies the BCD-C2 convergence requirements. In the following two sections, we respectively provide the detailed solving process of each subproblem and show that the subproblems satisfy the above convergence conditions.

C. Overview

This subsection presents the overall workflow of CoCaTS and explains how it operates in dynamic LEO satellite networks. Mainstream LEO satellite systems typically employ time synchronization and periodic link update mechanisms, such as Starlink (15 s) and OneWeb (11 s) [41]. Therefore, similar mechanism is adopted and CoCaTS is executed periodically. After each update period T , the large-scale LEO satellite network is partitioned into multiple small subnetworks using one of existing partitioning strategies [42, Sec. 3.2] [43, Sec. IV-A]. Then, for problem **P1**, the BCD method is applied to decompose it into three sub-processes for solution.

In specific, Algorithm 1 illustrates the detailed execution process for each update period. At first, the system update partition results, and then collect available LEO satellites in each partition area and active users in its coverage. Second, a manager should be elected among available LEO satellites in each partition area, and it updates caching decision $\mathbf{m}$ and user-satellite association $\mathbf{A}$ by executing Algorithms 2 and 4. Third, according to requests from active terrestrial users and real-time channel condition, the manager updates power allocation $\mathbf{P}$ at each frame duration d by executing Algorithm 3, where frame duration d is the minimum communication resource unit between LEO satellites and terrestrial users in time domain. During the update period, if a newly arrived user sends requests, a user-priority principle [44] is adopted, in which user-driven matching is performed based on the principle of optimizing users' QoS, to reduce excessive computational overhead. The

periodic update mechanism prevents severe degradation of the matching relationship under the user-priority principle. It should be noted that the terrestrial user experiences continuous service from LEO satellites in the same partition area, and the LEO satellite handover only happens at the end of each update period T . If an unexpected handover situation occurs, leading to a communication interruption, terrestrial users could experience temporary transmission failures. In this case, CoCaTS adopts a fallback strategy to guarantee service availability by retrieving the requested content through a bent-pipe satellite transmission scheme (as illustrated in Fig. 1) [1], [45].

IV. CACHING PLACEMENT

Caching placement decides which content files and how many coded blocks should be placed in on-board storage, which is a critical decision that contributes to the transmission cost. In practice, since the content files should be cached or updated in advance, the caching placement decision should be made before user-satellite association. In this section, we first consider a coded caching subproblem to obtain a global caching placement, and then user-satellite association subproblem could be further considered in the next section based on the solutions from caching placement.

A. Problem Decomposition

According to the joint optimization problem in **P1**, we first consider the caching placement subproblem. For a typical user u the subproblem can be formulated with given user-satellite association variable $\mathbf{a}$ and power allocation variable $\mathbf{P}$ as follows:

$$\max_{\mathbf{m}} \sum_{i \in \mathcal{F}} \tilde{p}_i O_u(\mathbf{m}) \quad (\text{P2})$$

$$s.t. \sum_i m_i \leq M, \forall i \in \mathcal{F}, \quad (\text{C2-1})$$

$$0 \leq m_i \leq c, \forall i \in \mathcal{F}, \quad (\text{C2-2})$$

$$m_i \in \mathbb{N}. \quad (\text{C2-3})$$

In specific, the problem can be interpreted as follows. Consider a knapsack with capacity of Mc blocks and $|\mathcal{F}|$ files consist of c blocks. The size of each block is the same, but the gain of each block varies according to the number of blocks as well as the index of file. Based on the proof process in [46, Theorem 1], we can also learn that subproblem **P2** is a multiple-choice knapsack problem, which is known to be NP-hard. In order to obtain the optimal caching placement, we first give some properties of the problem. Based on these properties, we devise a greedy but optimal algorithm as described in Algorithm 2.

To solve the caching placement subproblem, we provide the following lemma to give the optimal caching placement strategy.

Lemma 1: The optimal caching strategy for maximizing the offloaded traffic in **P2** satisfies that, $m_i > m_j$ for any given file popularity $p_i > p_j$.

Proof: We prove this lemma by using contradiction. Let $a < b$ and $a, b \in \mathcal{N}^+$. Assume that there are two files $f_1, f_2 \in \mathcal{F}$ and popularity $p_1 > p_2$. We further suppose that $\mathbf{m}^* = \{m_1 =$

$a, m_2 = b\}$ is an optimal caching strategy. Accordingly, we have another caching strategy $\mathbf{m}' = \{m_1 = b, m_2 = a\}$, which is obviously not the optimal caching strategy. Based on the above supposition, we have

$$O(\mathbf{m}^*) = K_u a + K_u b = K_u(a + b).$$

$$\sum_{i \in \mathcal{F}} \tilde{p}_i O(\mathbf{m}^*) = \tilde{p}_1 K_u a + \tilde{p}_2 K_u b = K_u(\tilde{p}_1 a + \tilde{p}_2 b).$$

Then we exchange the placement for file f_1, f_2 , and we have

$$O(\mathbf{m}') = K_u b + K_u a = K_u(b + a),$$

$$\sum_{i \in \mathcal{F}} \tilde{p}_i O(\mathbf{m}') = \tilde{p}_1 K_u b + \tilde{p}_2 K_u a = K_u(\tilde{p}_1 b + \tilde{p}_2 a),$$

where $K'_{a,b} = K_{1,u}(a) + K_{2,u}(b)$ and $K'_{b,a} = K'_{1,u}(b) + K'_{2,u}(a)$. Then, we have $K'_{a,b} = K'_{b,a}$. At last, we make the comparison between $O(\mathbf{m}^*)$ and $O(\mathbf{m}')$:

$$\begin{aligned} O(\mathbf{m}^*) - O(\mathbf{m}') &= K_u [\tilde{p}_1 a + \tilde{p}_2 b - \tilde{p}_1 b - \tilde{p}_2 a] = K_u [\tilde{p}_1(a - b) + \tilde{p}_2(b - a)] \\ &= K_u(\tilde{p}_1 - \tilde{p}_2)(a - b) < 0. \end{aligned}$$

From the assumption, we know that $p_1 > p_2$ and $a < b$. Then we can further get $O(\mathbf{m}^*) < O(\mathbf{m}')$, so $\mathbf{m}'$ represents a better block placement strategy, which leads to a contradiction to our previous assumption that the optimal block placement $\mathbf{m}^*$ should have $m_1 < m_2$ when $p_1 > p_2$. Thus, the optimal placement should satisfy $m_1 > m_2$ for any given file popularity $p_1 > p_2$. Lemma 1 is thus proved. $\square$

B. Algorithm Design

Lemma 1 provides a greedy strategy to find the optimal caching placement solution. Thus, we further design Algorithm 2 to find the optimal caching placement vector for all LEO satellites. The specific algorithm procedure can be described as follows. Lines 3 to 5 show the satellite sequentially calculating the overall performance improvement from caching one more file block for each file individually. Lines 6 to 15 describe the process of finding the file block with the highest benefits and then caching it in on-board storage. According to Lemma 1 and the iterative process of Algorithm 2, Algorithm 2 can find the optimal solution and thus subproblem **P2** has a unique solution, which satisfies the relaxed convergence conditions of BCD-C2.

Computational Complexity: The complexity of Algorithm 2 can be analyzed as follows. First, Lines 3 to 5 compute the performance improvement of O_u for each file when caching one additional block, and thus the iteration is executed $|\mathcal{F}|$ times. Second, the satellite selects the file block with the maximum current benefit to cache in lines 6 to 15, with the complexity of $\mathcal{O}(1)$. As described in line 2, this process is repeated for each file block until the on-board storage is full, resulting in an outer complexity of $\mathcal{O}(M)$. Therefore, the overall complexity of Algorithm 2 is $\mathcal{O}(M|\mathcal{F}|)$.

Algorithm 2: Caching Placement Algorithm.

Input: Cache Size M , number of original blocks c ;
Output: Caching placement vector $\mathbf{m}^*$;
1 **Initialize:** $\mathcal{I} = \emptyset$, $m_i = 0$, $\forall i \in \mathcal{F}$;
2 **while** $\sum_{i \in \mathcal{F}} m_i \leq M$ **do**
3 **for** $i \in \mathcal{F}$ **and** $m_i < c$ **do**
4 Compute $\delta_i = O_u(m_i + 1) - O_u(m_i)$;
5 **end**
6 **while true do**
7 Let $i' = \arg\max_{i \in \{\mathcal{F} - \mathcal{I}\}} \delta_i$;
8 **if** $m_{i'} < c$ **then**
9 Set $m_{i'} = m_{i'}, j + 1$;
10 **break**;
11 **end**
12 **else**
13 $\mathcal{I} = \mathcal{I} \cup \{i'\}$;
14 **end**
15 **end**
16 **end**

V. USER-SATELLITE ASSOCIATION

In this section, based on the caching placement decision, we further consider the user-satellite association subproblem with two coupled variables, i.e., user-satellite association $\mathbf{A}$ and corresponding power allocation $\mathbf{P}$. Since power-domain multiplexing technique is leveraged in the proposed NOMA-based cooperative transmission scheme, the decision for user-satellite association $\mathbf{A}$ has direct impact on power allocation $\mathbf{P}$ and SIC decoding process. Thus, we first consider single-user scenario to obtain the optimal power allocation strategy, and then extend this strategy to multi-user scenario.

A. Single User Scenario

For single-user scenario with given caching placement decision $\mathbf{m}$, i.e., $|\mathcal{U}| = 1$, problem **P1** can be decomposed into the following subproblem:

$$\max_{\mathbf{A}, \mathbf{P}} \sum_{i \in \mathcal{F}} \tilde{p}_i O(\mathbf{A}, \mathbf{P}) \quad (\text{P3})$$

$$s.t. \quad \|\mathbf{w}_{1,s}\| = 1, \forall s \in \mathcal{S}, \quad (\text{C3-1})$$

$$\sum_{s \in \mathcal{S}_u} \text{SINR}_{1,s} \geq \gamma_i, \quad (\text{C3-2})$$

$$P^{BB} + P^{RF} \leq P^{max}, s \in \mathcal{S}, i \in \mathcal{F}, \quad (\text{C3-3})$$

$$\sum_{s \in \mathcal{S}} a_{1,s} \leq \left\lceil \frac{c}{m_i} \right\rceil \leq K^{total}, \forall s \in \mathcal{S}, i \in \mathcal{F}. \quad (\text{C3-4})$$

For the typical user in subproblem **P3**, we consider that power constraint (C3-3) is always satisfied. In practical satellite system, the onboard transmitter usually integrates baseband and RF module together, and their energy consumption stays in watt-level [47], e.g., the typical power of Ka-band transmitter chip is 0.528/1.587 Watt (RX/TX) [48]. Thus, we focus on the variable power consumption of the RF module and consider

Algorithm 3: Optimal Power Allocation for **P3**.

Input: Number of cooperative transmission K_u ,
caching placement vector $\mathbf{m}$;
Output: Optimal power allocation $\mathbf{P}$;

- 1 **Initialize:** Sort $\mathcal{S}_u$ according their channel gain $\mathbf{h}_{u,k}$;
- 2 **for** $s = 1 : K_u$ **do**
- 3 **if** $\sum_s^{K_u} p_{u,s} < P^{max}$ **then**
- 4 $P_{u,s} = \frac{\gamma_i(\sum_{j \neq s} P_{u,j} |\mathbf{h}_{u,j} \mathbf{w}_{u,j}|^2 + N_0)}{|\mathbf{h}_{u,s} \mathbf{w}_{u,s}|^2}$;
- 5 $s = s + 1$;
- 6 **end**
- 7 **else**
- 8 **break**;
- 9 **end**
- 10 **end**

power consumption of the baseband module as a constant. Thus, constraint (C3-3) is to $P^{RF} \leq \tilde{P}^{max}$, where $P^{RF} = \mathbf{a}_u \mathbf{p}_{u,i} = \sum_{s=1}^{|\mathcal{S}|} a_{u,s} p_{u,s}$ and $\tilde{P}^{max} = P^{max} - P^{BB}$. Since the caching placement decision $\mathbf{m}$ is provided, the offloaded traffic in the objective is the same if the signal can be successfully decoded. In this case, the optimal transmission strategy is to minimize power consumption by deciding user-satellite association. Thus, the subproblem for single-user scenario can be transformed as follows:

$$\min_{\mathbf{A}, \mathbf{P}} \sum_{s=1}^{|\mathcal{S}|} a_{1,s} p_{1,s} \quad (\text{P4})$$

$$\text{s.t. } \|\mathbf{w}_{1,s}\| = 1, \forall s \in \mathcal{S}, \quad (\text{C4-1})$$

$$\sum_{s \in \mathcal{S}_u} SINR_{1,s} \geq \gamma_i, \quad (\text{C4-2})$$

$$\sum_{s \in \mathcal{S}} a_{1,s} \leq \left\lceil \frac{c}{m_i} \right\rceil \leq K^{total}, \forall s \in \mathcal{S}, i \in \mathcal{F}. \quad (\text{C4-3})$$

To solve the subproblem **P4**, we first give the following Lemmas to obtain the optimal power allocation strategy.

Lemma 2: For a typical user u requesting i -th content file with given caching placement vector $\mathbf{m}$, in order to minimize the transmission power, the optimal solution for NOMA-based cooperative transmission is to select satellites $\{s_1, s_2, \dots, s_{K_u}\}$ with K_u maximum channel gain following $\|\mathbf{h}_{u,1}\| > \|\mathbf{h}_{u,2}\| > \dots > \|\mathbf{h}_{u,K_u}\| > \dots > \|\mathbf{h}_{u,|\mathcal{S}|}\|$.

Proof: We prove this lemma by contradiction. Let the sequence of channel gain $\|\mathbf{h}_{u,1}\| > \|\mathbf{h}_{u,2}\| > \dots > \|\mathbf{h}_{u,|\mathcal{S}|}\|$. For $K_u = 2$, we assume that the optimal satellite selection scheme is $\{s_k, s_2\}$, $k \neq 1, 2$, and also the sub-optimal satellite selection scheme is $\{s_1, s_2\}$. For the optimal and the sub-optimal scheme, the transmission power consumption can be denoted by $\{p_{u,k}^*, p_{u,2}^*\}$ and $\{p_{u,1}^*, p_{u,2}^*\}$, respectively. To guarantee that user can decode the signal through SIC receiver, the minimum transmission power consumption can be given by (6) as

$$p_{u,k}^* = \frac{\gamma_i(P_{u,2}^* |\mathbf{h}_{u,2} \mathbf{w}_{u,2}|^2 + N_0)}{|\mathbf{h}_{u,k} \mathbf{w}_{u,k}|^2}, \quad (8)$$

$$p_{u,2}^* = \frac{\gamma_i N_0}{|\mathbf{h}_{u,2} \mathbf{w}_{u,2}|^2}. \quad (9)$$

Let $\mathbf{W} = \mathbf{w} \mathbf{w}^H$ and $\mathbf{H} = \mathbf{h}^H \mathbf{h}$. Here, $|\mathbf{h}_{u,2} \mathbf{w}_{u,2}|^2 = \text{Tr}(\mathbf{H}_{u,2} \mathbf{W}_{u,2})$, and $\mathbf{H}_{u,1} > \mathbf{H}_{u,2} > \mathbf{H}_{u,j}$ since $\|\mathbf{h}_1\| > \|\mathbf{h}_2\| > \|\mathbf{h}_k\|$, $k \neq 1, 2$. The diagonal element of $\mathbf{W}$ equals to 1 since $\|\mathbf{w}_{u,2}\| = 1$. In this case, we can have $|\mathbf{h}_{u,1} \mathbf{w}_{u,1}|^2 > |\mathbf{h}_{u,k} \mathbf{w}_{u,k}|^2$. Thus, the difference of power consumption between the optimal scheme and the sub-optimal can be calculated by

$$\begin{aligned} (p_{u,k}^* + p_{u,2}^*) - (p_{u,1}^* + p_{u,2}^*) &= p_{u,k}^* - p_{u,1}^* \\ &= (\gamma_i(p_{u,2} |\mathbf{h}_{u,2} \mathbf{w}_{u,2}|^2 + N_0)) \left(\frac{1}{|\mathbf{h}_{u,k} \mathbf{w}_{u,k}|^2} - \frac{1}{|\mathbf{h}_{u,1} \mathbf{w}_{u,1}|^2} \right) \\ &> 0, \end{aligned} \quad (10)$$

and thus $\{p_{u,1}^*, p_{u,2}^*\}$ represents a better power allocation with lower consumption, which leads to a contradiction to our previous assumption. For $K_u > 2$, we can have the similar process leads to a contradiction. Thus, the optimal power allocation should satisfy $\{s_1, s_2, \dots, s_{K_u}\}$ with $\|\mathbf{h}_{u,1}\| > \|\mathbf{h}_{u,2}\| > \dots > \|\mathbf{h}_{u,K_u}\| > \dots > \|\mathbf{h}_{u,|\mathcal{S}|}\|$. Lemma 2 is thus proved. $\square$

Lemma 3: For a typical user u requesting i -th content file with given caching placement vector $\mathbf{m}$ and user-satellite association vector $\mathbf{a}_u$, in order to minimize the transmission power, the optimal decoding sequence is to decode the signal from the best channel gain to the worst channel gain.

Proof: For any K_u , let sequence of channel gain satisfy $\|\mathbf{h}_{u,1}\| > \|\mathbf{h}_{u,2}\| > \dots > \|\mathbf{h}_{u,K_u}\|$. If we decode the signal from the best channel gain to the worst channel gain, then we can have minimum transmission power p_n^* and p_{n+1}^* for the n -th and $(n+1)$ -th decoding layer as

$$\begin{aligned} p_{u,n}^* &= \frac{\gamma_i \left(\sum_{s=n+2}^{K_u} P_{u,s} |\mathbf{h}_{u,s} \mathbf{w}_{u,s}|^2 + N_0 \right)}{p_{u,n+1}^* |\mathbf{h}_{u,n+1} \mathbf{w}_{u,n+1}|^2 + N_0}, \end{aligned} \quad (11)$$

$$p_{u,n+1}^* = \frac{\gamma_i \left(\sum_{s=n+2}^{K_u} P_{u,s} |\mathbf{h}_{u,s} \mathbf{w}_{u,s}|^2 + N_0 \right)}{|\mathbf{h}_{u,n+1} \mathbf{w}_{u,n+1}|^2}, \quad (12)$$

where $1 \leq n \leq K_u$ is satisfied. If we decode the signal in any other order through SIC receiver, i.e., decoding the $(n+1)$ -th layer at first, then decoding the n -th layer. In this way, we can have minimum transmission power $p'_{u,n}$ and $p'_{u,n+1}$ for the n -th and $(n+1)$ -th decoding layer as

$$p'_{u,n} = \frac{\gamma_i \left(\sum_{s=n+1}^{K_u} P_{u,s} |\mathbf{h}_{u,s} \mathbf{w}_{u,s}|^2 + N_0 \right)}{|\mathbf{h}_{u,n} \mathbf{w}_{u,n}|^2}, \quad (13)$$

$$\begin{aligned} p'_{u,n+1} &= \frac{\gamma_i \left(\sum_{s=n+2}^{K_u} P_{u,s} |\mathbf{h}_{u,s} \mathbf{w}_{u,s}|^2 + P'_{u,n} |\mathbf{h}_{u,n} \mathbf{w}_{u,n}|^2 + N_0 \right)}{|\mathbf{h}_{u,n+1} \mathbf{w}_{u,n+1}|^2}. \end{aligned} \quad (14)$$

Thus, the difference of power consumption between two decoding scheme can be calculated by

$$\begin{aligned}
 & (p_{u,n}^* + p_{u,n+1}^*) - (p_{u,n}' + p_{u,n+1}') \\
 &= \frac{\gamma_i p_{u,n+1}^* |\mathbf{h}_{u,n+1} \mathbf{w}_{u,n+1}|^2}{|\mathbf{h}_{u,n} \mathbf{w}_{u,n}|^2} - \frac{\gamma_i p_{u,n}' |\mathbf{h}_{u,n} \mathbf{w}_{u,n}|^2}{|\mathbf{h}_{u,n+1} \mathbf{w}_{u,n+1}|^2} \\
 &= \frac{\gamma_i N_0}{|\mathbf{h}_{u,n} \mathbf{w}_{u,n}|^2} - \frac{\gamma_i N_0}{|\mathbf{h}_{u,n+1} \mathbf{w}_{u,n+1}|^2} < 0, \quad (15)
 \end{aligned}$$

which implies that decoding from the best channel gain to the worst channel gain always has minimum power consumption.

Lemma 3 is thus proved. $\square$

Corollary 1: In order to minimize the transmission power for a user u , it is necessary to select satellites with K_u maximum channel gain, and allocate transmission power as given in (6) following the order from the best channel gain to the worst channel gain, i.e., $\|\mathbf{h}_{u,1}\| > \|\mathbf{h}_{u,2}\| > \dots > \|\mathbf{h}_{u,K_u}\|$.

According to Corollary 1, subproblem **P4** exhibits monotonicity with respect to channel gain, and thus we design Algorithm 3 to obtain the optimal power allocation for subproblem **P4**. In specific, lines 3 to 9 describe the process of sequentially allocating transmission power for user u based on the order of channel gain, until the power condition for successful decoding is met. Consequently, subproblem **P4** has a unique solution by using Algorithm 3, and its constraint set is a Cartesian product of closed convex sets, which satisfies the convergence conditions of BCD-C1 [37]. Meanwhile, the time complexity of Algorithm 3 is $\mathcal{O}(K_u)$.

B. Multi-User Scenario

For multi-user scenario with given caching placement vector m , the joint optimization problem **P1** can be decomposed as follows:

$$\max_{\mathbf{A}, \mathbf{P}} \sum_{i \in \mathcal{F}} \sum_{u \in \mathcal{U}} \tilde{p}_i O_u(\mathbf{A}, \mathbf{P}) \quad (\text{P5})$$

$$s.t. \quad \|\mathbf{w}_{u,k}\| = 1, \forall u \in \mathcal{U}, \forall s \in \mathcal{S}, \quad (\text{C5-1})$$

$$\sum_{s \in \mathcal{S}_u} SINR_{u,s} \geq \gamma_i, \forall a_{u,s} = 1, \quad (\text{C5-2})$$

$$P^{BB} + P^{RF} \leq P^{max}, \forall u \in \mathcal{U}, s \in \mathcal{S}, i \in \mathcal{F}, \quad (\text{C5-3})$$

$$K_u \leq \left\lceil \frac{c}{m_i} \right\rceil \leq K^{total}, \forall u \in \mathcal{U}, i \in \mathcal{F}, \quad (\text{C5-4})$$

$$a_{u,s} \in \{0, 1\}, p_{u,s,i} \in \mathbb{R}. \quad (\text{C5-5})$$

Similar to subproblems **P3** and **P4** in single-user scenario, user-satellite association and power allocation are coupled. Since subproblem **P5** is still a MINLP problem, which is NP-hard and can hardly derive a global optimal solution with polynomial-time complexity. To address this complicated subproblem, a matching based technique is preferred, which we can be explained in the following two aspects. On the one hand, user-satellite association is inherently a matching process between terrestrial users and LEO satellites, combining with the optimal power strategy proved in Section V-A, an effective solution can be

easily obtained. On the other hand, for ultra-dense LEO satellite networks comprising thousands of LEO satellites in a mega constellation, at least dozens of satellites can be reached for a typical user in an arbitrary geographical location. In this case, a close-to-optimal yet low-complexity solution is preferred. Thus, we propose a multiple-to-multiple two-side matching model with externalities [49], [50] to solve the association problem for terrestrial users and LEO satellites. Note that once the association process completes, we can derive the power allocation solution since the problem can degenerate into a simple power allocation problem in single user scenario. In the next, we consider the many-to-many matching model between LEO satellites and terrestrial users, which is defined as follows.

C. Many-to-Many Matching Solution

1) *Many-to-Many Matching Model:* For convenience, we denote (u, s) as a matching pair if user u is served by s -th satellite. At first, a matching is an assignment of satellites in $\mathcal{S}$ to users in $\mathcal{U}$, which can be defined as follows.

Definition 1: Given two disjoint sets, $\mathcal{S} = \{1, 2, \dots, |\mathcal{S}|\}$ of the satellites, $\mathcal{U} = \{1, 2, \dots, |\mathcal{U}|\}$ of the terrestrial users, a many-to-many matching Φ is a mapping from the set $\mathcal{S} \cup \mathcal{U}$ into the set of all subsets of $\mathcal{S} \cup \mathcal{U}$ such that for every $s \in \mathcal{S}, u \in \mathcal{U}$:

1. $\Phi(u) \subseteq \mathcal{S}, \Phi(s) \subseteq \mathcal{U}$.
2. $0 \leq |\Phi(s)| \leq K_s^{max}, 0 \leq |\Phi(u)| \leq K_u^{max}$.
3. $s \in \Phi(u) \leftrightarrow u \in \Phi(s)$.

Note that condition 1 denotes that each player is matched with a subset of other player, condition 2 limits that each user can be assigned up to K_u satellites and each satellite can serve K_s' users at most, condition 3 indicates the set of players in the match pairing is symmetrical.

Next, we define the preference value of satellites on the set of users under the matching state as the power consumption, and the preference value of users on the set of satellites under the matching state as the channel quality. The preference lists are based on the channel quality and power consumption, respectively. They can be denoted as:

$$PF_{SAT} = [PF_{SAT}(1), \dots, PF_{SAT}(u), \dots, PF_{SAT}(|\mathcal{U}|)], \quad (16)$$

$$PF_{USER} = [PF_{USER}(1), \dots, PF_{USER}(s), \dots, PF_{USER}(|\mathcal{S}|)], \quad (17)$$

where PF_{SAT} and PF_{USER} are the preference lists of satellites and users, respectively. In the following, we further provide the definition of a matching game.

Definition 2: For a given matching game, two preference relations $\succ_u, \succ_s$ allowing each player $u \in \mathcal{U}, s \in \mathcal{S}$ to build preferences over one another, i.e., to rank, respectively, the players in $\mathcal{U}$ and $\mathcal{S}$. For any user u , a preference relation $\succ_u$ is defined over the set of satellites $\mathcal{S}$, and two matchings $\Phi, \Phi' \in \mathcal{U} \times \mathcal{S}$, $u \in \Phi(s), u' \in \Phi'(s')$:

$$(u, s) \succ_u (u', s') \Leftrightarrow U_u(\Phi) > U_{u'}(\Phi'). \quad (18)$$

For any satellite s , a preference relation $\succ_s$ is defined over the set of users $\mathcal{S}$, and two matchings $\Phi, \Phi' \in \mathcal{U} \times \mathcal{S}$, $s \in \Phi(u), s' \in$

$\Phi(u')$:

$$(u, s) \succ_s (u', s') \Leftrightarrow U_s(\Phi) > U_{s'}(\Phi'). \quad (19)$$

Since LEO satellites share the same resource (i.e., spectrum) to serve the same user, and the terrestrial user adopts a SIC receiver to decode the signal from multiple satellites, the power allocation for each satellite is relevant to other satellites serving the same user. Thus, each satellite not only cares which user it is matched with, but also the set of other satellites matching with the same user. For this reason, the aforementioned matching game is called the many-to-many game with externalities [51], [52] or peer effect [53], where each player has a dynamic preference list over the opposite set of players. This is different from the conventional matching games in which players have fixed preference lists [51].

Remark 1: The proposed matching has externalities, where the preference list of satellites not only depend on the channel quality of terrestrial users that they matched with, but also on the other satellites matched to the same user.

Due to the externalities, the many-to-many matching model considered in this paper is more complicated than the conventional two-sided matching models. Under traditional definition of stable matching in [52], there is no guarantee that a stable matching exists even in many-to-one matchings. Because of the lack of substitutability, traditional solution, such as deferred acceptance algorithm, cannot be applied to this model any more. To solve this matching problem, we introduce the concept of swap matching and propose a swap matching algorithm.

2) *User-Satellite Matching Algorithm:* For the proposed many-to-many two-side matching model, our goal is to find a stable matching. With the notations of swap matching and the definition of stability, we propose a swap matching algorithm as described in Algorithm 4. This algorithm is an extension of the many-to-one matching algorithm proposed in [54] with constraints. At first, we introduce the concept of swap matching as follows.

Definition 3: Given a matching Φ , two matching pairs (u, s) and (u', s') with $u \in \Phi(s)$, $u' \in \Phi(s')$, $s \notin \Phi(u')$, $s' \notin \Phi(u)$, a swap matching is defined as:

$$\Phi_{u,s}^{u',s'} = \{\Phi \setminus \{(u, s), (u', s')\} \cup \{(u', s), (u, s')\}\}. \quad (20)$$

As stated in [52], there is no longer a guarantee that a traditional “pairwise-stability” exists when players care about more than their own matching, and if a stable matching does exist, it can be computationally difficult to find.

Definition 4: For a given matching Φ , u and u' are matched in Φ . When $s \in \Phi(u)$ and $s' \in \Phi(u')$, if the following conditions are satisfied:

1. $\forall x \in \{u, u', s, s'\}, (\Phi_{u,s}^{u',s'}(x), x) \succeq (\Phi(x), x)$,
2. $\exists x \in \{u, u', s, s'\}, (\Phi_{u,s}^{u',s'}(x), x) \succ (\Phi(x), x)$,

then the pair $\Phi_{u,s}^{u',s'}$ is called a **swap-blocking pair** in the matching Φ and the swap matching is approved.

The condition 1 implies that the utilities of all the involved players should not be reduced after the swap operation between the swap-blocking pair. The condition 2 indicates that at least

Algorithm 4: User-satellite Matching Algorithm.

Input: Caching placement $\mathbf{m}$;
Output: User-satellite matching Φ ;

- 1 **Initialize:**
- 2 • Calculate PF_{SAT} according to power consumption;
- 3 • Calculate PF_{USER} according to channel quality;
- 4 • Calculate $K_u^{max} = \lceil \frac{c}{m_i} \rceil, K_s^{max} = \lfloor \frac{P^{max}}{P_{u,min}} \rfloor$;
- 5 • Users and satellites are randomly matched with each other subject to constraints (users try to connect to satellites as much as possible).
- 6 **while** $label == true$ **do**
- 7 $label = false$
- 8 Update utility based on the current Φ ;
- 9 Update PF_{SAT} and PF_{USER} ;
- 10 **for** $u \in \mathcal{U}$ **do**
- 11 **for** $s^* \notin PF_{USER}$ **do**
- 12 **if** (u, s^*) forms a swap-blocking **then**
- 13 User u sends a request to s^* ;
- 14 **if** (u, s^*) is approved, i.e., constraints are satisfied **then**
- 15 u_i swaps its match s_p with u_j for s_q
- 16 $\Phi \leftarrow \Phi \setminus (u_i, s_p) \setminus (u_j, s_q)$
- 17 $\Phi \leftarrow \Phi \cup (u_i, s_q) \cup (u_j, s_p)$
- 18 $label = true$;
- 19 **break**;
- 20 **end**
- 21 **else**
- 22 keep the current matching state;
- 23 **end**
- 24 **end**
- 25 **end**
- 26 **end**
- 27 **end**
- 28 Output the final matching result Φ ;

one of the players' utilities is increased after the swap operation between the swap-blocking pair.

Definition 5: A matching Φ is two-sided swap-stable if and only if there does not exist a swap-blocking pair.

To find a stable matching for the user-satellite association problem in P5, we propose Algorithm 4 and give remarks on its complexity, convergence, and optimality.

Complexity: Lines 10 to 14 describe querying all combinations that can perform swap matching. First, all users u_i and all satellites s_q not connected to the selected user are traversed, with a complexity of $\mathcal{O}(K_u \mathcal{U})$. Then, it searches for satellite s_p connected to user u_i and user u_j connected to satellite s_q , where u_j is not connected to s_p , which also has a complexity of $\mathcal{O}(K_u \mathcal{U})$. Lines 15 to 19 describe performing a swap matching if a swap-blocking pair occurs, and then ending the query. To achieve Nash equilibrium, only one swap matching is performed per query. The complexity of individual operations is $\mathcal{O}(1)$. Therefore, the overall complexity of Algorithm 4 is $\mathcal{O}(K_u^2 \mathcal{U}^2)$. The complexity grows with the fourth power of the area of the

partitioned region. To reduce computation time and regional heterogeneity, the LEO satellite network should be divided into multiple regions for independent matching during practical deployment.

Stability and Convergence: The potential function ϕ of the swap matching can be defined as

$$\phi = \sum_{u \in \mathcal{U}} \sum_{s \in \mathcal{S}} F(u, s), \quad (21)$$

where F is the connection utility function, i.e., the sum of the user and satellite's utilities at both ends of the link. After each swap, ϕ strictly increases and satisfies three conditions: the number of users and satellites is finite, each connection has a bounded function utility F , and ϕ has an upper bound (i.e., $F_{\max} \times \text{number of connections}$). Therefore, ϕ is guaranteed to converge within a finite number of swaps:

$$\exists Ti < \infty, \quad \phi^{Ti+1} = \phi^{Ti}. \quad (22)$$

Algorithm 4 solves subproblem **P5** with a unique solution and satisfies the relaxed convergence conditions of BCD-C2.

Optimality: Algorithm 4 can be regarded as a locally improved greedy search algorithm that converges to a stable matching state within a finite number of steps. Its termination condition is that no bilateral connection swap satisfying the local improvement criterion exists, indicating that the system reaches a locally optimal state. Since the system only permits swap operations, the search space is confined to adjacent matching states, preventing the algorithm from escaping local optima. Therefore, while Algorithm 4 guarantees the stability of the matching, it sacrifices global efficiency.

Overhead: After each connection refresh in the LEO satellite network, Algorithm 4 be re-executed. Satellites inside a partition send their load state, link state, and channel state to the manager satellite. The manager executes Algorithm 4, obtains the new matching, and then distributes the results to the satellites. In an $N \times N$ network, the number of transmissions required to collect states or to distribute results can be computed using the Manhattan distance. When the manager satellite is located at a corner, the maximum number of transmissions is required, which is calculated as $N^3 - N^2$. Satellite state information or matching results are encapsulated in IP packets. Without loss of generality, the state packet size is considered as $30B + 8B/Link$ (8 bytes per link) to record satellite load state and each link state. The result packet size is considered as $26B + 3B/File + 2B/Satellite$ (3 bytes per file and 2 bytes per satellite) to record cache and matching results. In a typical 4×4 partition, the maximum number of transmissions is 48, and then the reconfiguration overhead (i.e., signaling costs) for one re-execution of Algorithm 4 does not exceed 29 KB.

VI. NUMERICAL EVALUATION

In this section, we conduct comprehensive evaluations of the proposed CoCaTS compared to baselines. We begin by presenting the simulation model, parameter settings, and baseline schemes for LEO satellite networks. Then, we assess the convergence properties of CoCaTS. Finally, we conduct

TABLE II
EVALUATION PARAMETERS

Parameter	Value
Number of users	100
Number of satellites	16 (4×4)
Number of files	100
Block of files	9 ($8 + 1$)
Size of entire-files	1GB
User decoding threshold (γ)	0.5874
Satellite altitude	780km
Satellite EIRP	36.7dBw
P^{BB}	2.0058 dBW
Satellite link	25
Satellite storage space	10GB
Satellite coverage radius	500km
Center frequency	13.5GHz
Subcarrier bandwidth	10MHz
Power spectral density of noise	-174dBm/Hz

EIRP: Effective Isotropic Radiated Power.

extensive evaluations of various performance metrics under dynamic network scenarios.

A. Simulation Setup

In the real world, large LEO satellite networks are generally managed through partitioning [2], [13], which helps reduce overall complexity and shorten control loops. We conduct simulation experiments within one such partitioned area. Within a $100\text{km} \times 100\text{km}$ range, a certain number of users and satellites are generated. Satellites are uniformly distributed in a checkerboard pattern within the area, while users are randomly generated. Each user is guaranteed to be able to connect to at least one satellite. All evaluations are performed on a PC with 16-core Intel i5-13400 CPUs (2.5 GHz), 32 GB RAM, and Windows 11 OS.

To make the experimental results more realistic, we refer to the actual parameters of real-world LEO satellite networks [2], [5], [55], as well as the channel model and authoritative channel parameter settings [56]. The specific parameters are shown in Table II. Parameters modified for different scenarios are emphasized separately.

The comparison schemes we adopt are as follows:

- *CoCaTS*: The proposed cooperative coded caching-enabled transmission scheme.
- *UP (User-preferred link)*: Matching is performed based on the principle of optimizing the users' QoS, inspired by [44].
- *SP (Satellite-preferred link)*: Matching is performed based on the principle of minimizing the satellite's power consumption, inspired by [25].
- *Baseline 1 (Single-link and no-caching, shown in Fig. 1.1)*: The basic transmission scheme used in real LEO satellite networks [5]. LEO satellites and terrestrial users are connected via single links, and no content is cached on the satellites.

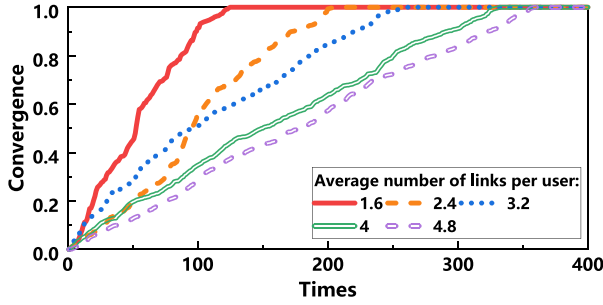


Fig. 3. Normalized convergence of swap matching in different number of links per user (K_u).

- *Baseline 2 (Cooperative entire-file cache, shown in Fig. 1.3):* Satellite regional cooperative caching is used, but files are cached as a whole. Single-link transmission is implemented [13].
- *Baseline 3 (Random cache, shown in Fig. 1.2):* This scheme uses a random caching strategy for file blocks.
- *Baseline 4 (StarCDN):* This scheme adopts K buckets with consistent hashing for cooperative caching, and employs inter-satellite routing [42].

B. Performance Evaluation

We first evaluate the convergence of the matching algorithm, as its convergence determines the stability of transmission. Fig. 3 shows the convergence and convergence speed under different numbers of LEO satellite links. Convergence is measured by the normalized total reduction in link power consumption. In all scenarios, the LEO satellite network converges to a Nash equilibrium, after which no further swap-blocking pairs exist. Once the matching stabilizes, no better or worse swaps can occur, and the connection utility remains constant. As the number of LEO satellite links increases, the matching process becomes more complex, requiring more iterations to converge. Under the default setting of 25 links (Avg. link: 4), the algorithm takes approximately 330 matching operations to reach convergence.

We consider that terrestrial users randomly send requests and then download content files from resource-consuming application server according to the same Zipf distribution in (1). The LEO satellite network is continuously in motion, and terrestrial users dynamically connect to and disconnect from the network. The number of active users and the requested files randomly vary at each timestamp. Accordingly, we set 50 timestamps to repeatedly evaluate the performance of each algorithm under dynamic network conditions. Meanwhile, we conduct extensive experimental evaluations on various metrics, including offloaded traffic, transmission download time, on-board overhead in terms of transmission power and storage, and cache hit rate, under different influencing parameters.

In Fig. 4, we evaluate the performance for different numbers of service users. Given the partition range, the number of users is equivalent to the user density. Therefore, the performance of CoCaTS under different user densities, i.e., load capacity, is explored. Fig. 4(a) shows the average file download time for users. With the growing number of users, increased satellite

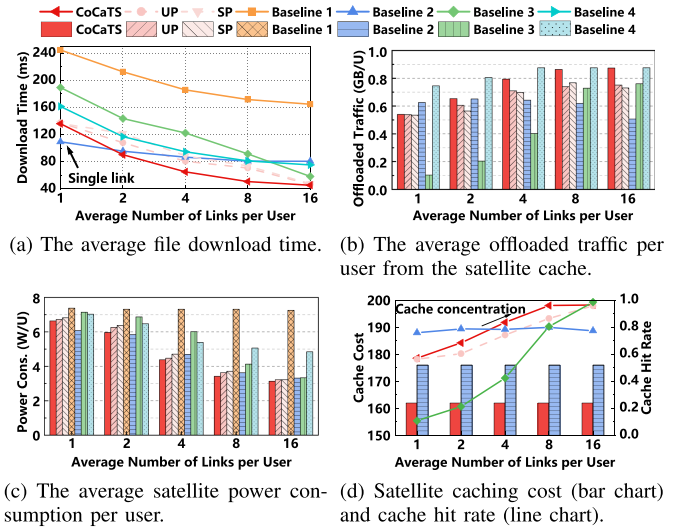


Fig. 4. The average performance of different caching and transmission algorithms under varying numbers of users (400, 200, 100, 50, and 25). (Users are randomly distributed, Zipf parameter $\theta = 1$).

load and reduced average link number per user collectively lead to longer download times and gradually diminish the benefits of multi-satellite cooperative transmission. When the average number of links per user falls below 2, some users can only retrieve files through a single link. Due to the additional cost of decoupled encoding caching, the file download time of CoCaTS exceeds that of Baseline 2. However, when sufficient links are available (more than 4), CoCaTS reduces the average file transmission time by at least 61.63%, 28.49%, 46.41%, and 30.5% compared to Baseline 1, 2, 3, and 4 respectively. Fig. 4(b) shows the offloaded traffic directly from the satellite cache for different algorithms. Each user requests a file only once. Similar to Fig. 4(b), when the user number is high, the effective offloaded traffic of CoCaTS becomes lower than that of Baseline 2 due to the additional encoded blocks. When the user number is not overloaded (such as 100), the offloaded traffic of CoCaTS increases by 28.5% and 105.55% compared to Baseline 2 and Baseline 3, respectively. Baseline 4 adopts cooperative caching and inter-satellite routing, which efficiently utilizes storage and Inter-Satellite Link (ISL) bandwidth and therefore maintains relatively high offloaded traffic. However, when the number of available links becomes sufficient (more than 8), its performance becomes comparable to that of CoCaTS. Compared to the variant schemes UP and SP, CoCaTS also achieves improvements of 8.12% and 12.07% in offloaded traffic due to its optimized matching strategy.

Fig. 4(c) shows the satellite power consumption of each algorithm. Compared with Baseline 1, CoCaTS significantly reduces power consumption by at least 25.34%. When the user number is high, due to the need to transmit additional coded blocks, CoCaTS exhibits slightly higher power consumption than Baseline 2 (about 4.43%), which does not require coding. When the user number is low, CoCaTS consumes approximately 33.63% less power than Baseline 2, thanks to shorter transmission paths and an optimized power allocation strategy.

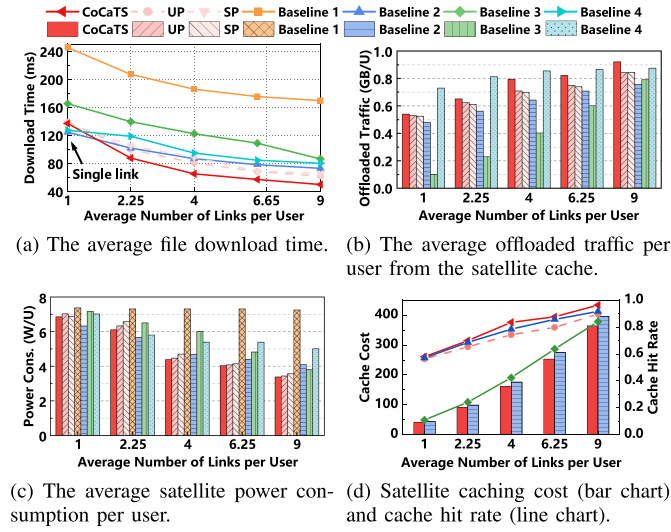


Fig. 5. The average performance of different caching and transmission algorithms with varying numbers of satellites (4, 9, 16, 25, and 36). (Satellites are evenly distributed in a checkerboard pattern. Zipf $\theta = 1$).

Compared with Baseline 4, which relies on inter-satellite routing and transmits content over multiple ISLs, CoCaTS reduces power consumption by an average of 40.81%. Benefiting from a more stable matching algorithm, CoCaTS also achieves about 6.25% and 4.04% lower power consumption compared to UP and SP. Fig. 4(d) shows the caching cost and cache hit rate. CoCaTS enable more flexible cache-transmission patterns and do not require entire file caching, resulting in lower overall caching cost than Baseline 2. The coded blocks are distributed across multiple satellites, yielding a higher overall cache hit rate. However, as the number of users increases, each user connects to fewer satellites. To ensure that users achieving cache hits can successfully decode, satellites must cache more coded blocks of the same file, reducing the diversity of cached files. This leads to a lower cache hit rate compared to Baseline 2 under high user load. In addition, the internal matching competition in UP and SP results in less stable connections, which further reduces cache hit rate compared to CoCaTS.

In Fig. 5, we evaluate the performance of CoCaTS under different satellite numbers (densities). As the number of satellites increases, the overall cache space and bandwidth increase, leading to a performance improvement for all schemes. It is noteworthy that CoCaTS employs a swap matching algorithm, which achieves more stable matching under dense satellite connections. Therefore, CoCaTS obtains higher performance gains compared to other schemes. Compared with Baseline 1, CoCaTS reduces the download time by at least 44.47% and the power consumption by an average of 24.52%. Similar to Fig. 4, when the number of satellites is small, the limited number of links leads to a longer download time for CoCaTS compared to Baseline 2 and 4. However, by resolving user contention, CoCaTS achieves 18.48% higher offloaded traffic than Baseline 2. When the number of satellites exceeds 16, CoCaTS reduces the download time by at least 28.49% and power consumption by at least 22.71% compared to Baseline 2, while also achieving

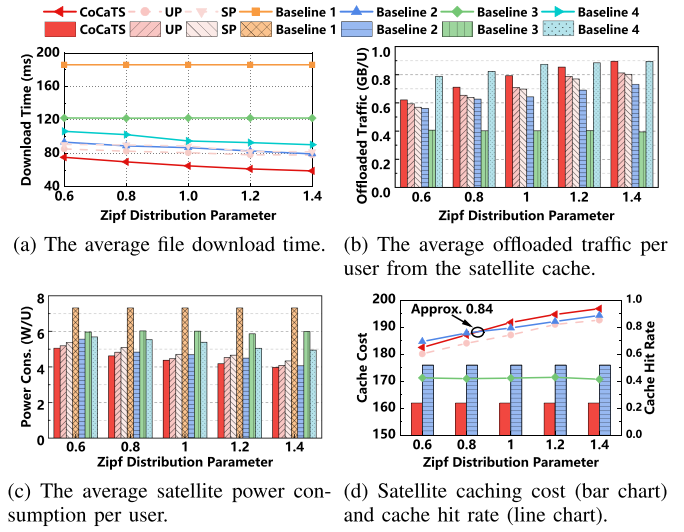


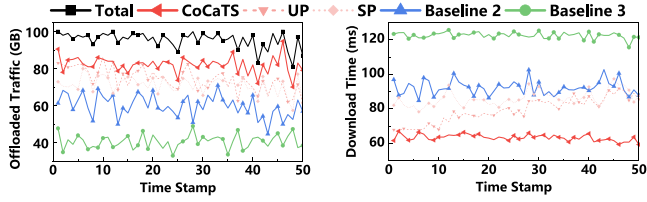
Fig. 6. The average performance of different caching and transmission algorithms under different file popularity distributions (Zipf θ).

a higher cache hit rate with lower caching cost. Meanwhile, compared with Baseline 4, CoCaTS reduces the download time by at least 26.84%, lowers power consumption by 31.13%, and maintains a comparable level of offloaded traffic. The abundant cache space weakens the impact of caching strategies, and the performance of Baseline 3 rapidly improves, approaching that of CoCaTS.

In Fig. 6, we evaluate the performance of CoCaTS under different file popularity distributions. As θ increases, the file popularity distribution becomes more concentrated, and the performance of cache-based schemes improves. Compared with Baseline 1, 2, 3, and 4, CoCaTS reduces the download time by at least 56.41%, 26.99%, 39.6%, and 29.59%, respectively, and increases the offloaded traffic by at least 28.5% (2) and 67.81% (3), respectively. Baseline 2, which caches entire files and relies on single-link transmission, always caches the most popular files. However, as the file popularity distribution becomes more concentrated, the satellites caching these popular files experience increased link pressure. Users who fail to connect must request files from servers via other satellites, resulting in a lower cache hit rate and higher transmission cost for Baseline 2 compared to CoCaTS.

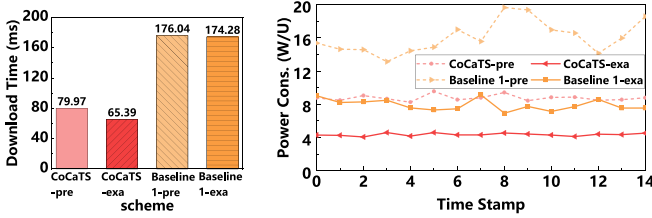
From Figs. 4, 5 and 6, we observe that simple encoding-decoupled caching combined with swap matching cannot outperform schemes that cache entire files based on popularity. However, when combining encoding-decoupled caching and popularity-based caching with cooperative transmission, even if the matching strategy is slightly worse, the performance is still higher than that of single-link transmission most of the time. In any case, these new schemes perform better than the baseline currently used in practice.

In Fig. 7, we evaluate the time-varying performance of different algorithms in a dynamic LEO satellite network scenario. CoCaTS remains stable and adapts well to the dynamic user request. Initially, UP's performance is close to CoCaTS, but over time, new users focus solely on their own QoS, leading to intense



(a) The total offloaded traffic from the satellite cache. (b) The average file download time.

Fig. 7. The performance of different algorithms in a dynamic LEO satellite network scenario. The time stamp indicates the update period. The number of users and the files requested by users change over time (Zipf $\theta = 1$).



(a) The average file download time.

(b) The average power consumption.

Fig. 8. The performance of different algorithms under dynamic channel conditions. The time stamp indicates that the channel conditions are updated every frame (Zipf $\theta = 1$).

competition between different users and with already connected users. This results in severe load imbalance and deteriorating connection relationships. SP faces the same issue, causing the connection relationships to degrade and deviate further from the optimal solution. Ultimately, all algorithms, except CoCaTS, converge to lower performance levels than CoCaTS. Regarding file download time, CoCaTS is significantly better than the other schemes. Initially, UP outperforms SP, but as connection relationships deteriorate, both performances decline and become almost identical. In addition, the performance of Baseline 2 is close to that of SP, while Baseline 3 exhibits the worst performance due to its caching strategy. In summary, CoCaTS demonstrates excellent stability and convergence, effectively supporting traffic offloading for a large number of users.

In Section III, CoCaTS assumes accurate CSI and ideal SIC decoding as the basis for theoretical analysis. However, in a practical LEO environment, the acquisition of CSI relies on channel state measurement. For example, ECOCSI [57], a CSI prediction scheme for satellite-terrestrial networks without terrestrial device position and LEO ephemeris, can achieve a rather low prediction error under various LEO environments with the maximum error CSI_{er} being approximately 10%. In this case, such errors can introduce a performance gap between theoretical performance and practical deployments. Thus, here, we further adopt the method described in ITU-R P.618 [58] to predict CSI and add redundant transmission power to ensure that signals do not fail to decode due to channel fluctuations and prediction error. Specifically, at 13.5 GHz, a redundancy margin of approximately 3 dB ensures that about 99% of signals are successfully decoded. In the experiment shown in Fig. 8, we evaluate the impact of dynamic channel conditions on the performance of CoCaTS, the channel conditions are regenerated in each frame

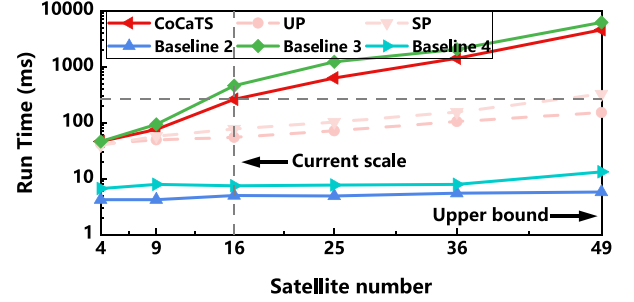


Fig. 9. The runtime complexity of different algorithms under various network scales (Number of satellites in a partition).

using the model in Section III-A-2), and the transmission power is reallocated accordingly. In Fig. 8, CoCaTS-pre indicates the performance of CoCaTS using predicted CSI, and CoCaTS-exa indicates the performance of CoCaTS using accurate and exact CSI. In specific, when predicted CSI is used, the satellite power consumption increases by nearly one time. Even so, CoCaTS-pre remains only slightly higher than the theoretical minimum consumption of the current transmission scheme and is still far lower than its practical consumption. Due to limited satellite power, the satellite cannot serve all connected users at the maximum rate and files need to wait in a transmission queue, which increases the file download time by approximately 22.3% compared with the theoretical value, but it remains significantly lower than that of the baseline.

C. Computational Feasibility

This subsection further discusses how a simple and effective partitioning strategy can ensure the computational feasibility of Algorithm 2 and 4 in practical ultra-dense LEO satellite networks. The partitioning of LEO satellite networks can be based on ground regions [2], network density [13], or network topology [42]. Current mainstream LEO satellite networks, such as Starlink, adopt a four-link pattern, where each satellite connects to two adjacent satellites in the same orbit and two in neighboring orbits (east-west). Based on the network topology, partitioning the network into $N \times N$ grids is a practical and manageable approach [42]. Most ground regions can receive signals from more than 10 satellites simultaneously. When Starlink's Gen 2 constellation (approximately 30,000 satellites [59]) is fully deployed, the terrestrial users can receive signals from up to 44 satellites theoretically. In practical implementations, the partitioning strategy used in StarCDN [42] is first applied to partition the network into manageable subnetworks, followed by the execution of CoCaTS, which ensures computational feasibility.

Fig. 9 presents the runtime complexity of different algorithms under various network scales. Under the current scale of LEO satellite networks, CoCaTS requires only a little over 200 ms to complete. Compared to the typical update period $T = 15$ s in the simulation, the runtime of the algorithm only requires 1.3% of total time duration. CoCaTS and Baseline 3 employ more sophisticated matching algorithms and therefore exhibit slightly higher runtime than the other schemes. The runtime

complexity is primarily related to K_u , the average number of satellites connected to a user. Since K_u is determined by the partition size and the number of users within that partition, it has no direct relationship with the overall network scale. Therefore, the runtime can be guaranteed by adjusting the partition size.

VII. CONCLUSION

In this paper, considering the scenario of ultra-dense satellite deployment, we designed a cooperative caching-enabled transmission scheme to degrade the cost and enhance the performance while serving resource-consuming applications. Specifically, to maximize users' offloading traffic, we first formulated a joint optimization problem for caching placement, power allocation, and user-satellite association. We then decomposed this NP-hard problem into subproblems and designed efficient algorithms to obtain solution for each subproblem. Finally, the numerical results show that, compared to existing solutions, our scheme shows superior performance in terms of download time, offloaded traffic, power consumption, cache cost, and cache hit rate. Specifically, the proposed scheme, i.e., CoCaTS, shows increasing benefits with the increase of available LEO satellites compared to existing ones. While the ongoing construction of mega constellation, CoCaTS becomes attractive potential in future ultra-dense LEO satellite networks.

REFERENCES

- [1] X. Cao and X. Zhang, "SaTCP: Link-layer informed TCP adaptation for higuuhhhly dynamic LEO satellite networks," in *Proc. IEEE INFOCOM 2023-IEEE Conf. Comput. Commun.*, 2023, pp. 1–10.
- [2] J. Tang et al., "Cooperative caching in satellite-terrestrial integrated networks: A region features aware approach," *IEEE Trans. Veh. Technol.*, vol. 73, no. 7, pp. 10602–10616, Jul. 2024.
- [3] Y. Li et al., "A networking perspective on Starlink's self-driving LEO mega-constellation," in *Proc. ACM Int. Conf. Mobile Comput. Netw.*, 2023, pp. 1–16.
- [4] J. Li, H. Lu, K. Xue, and Y. Zhang, "Temporal netgrid model-based dynamic routing in large-scale small satellite networks," *IEEE Trans. Veh. Technol.*, vol. 68, no. 6, pp. 6009–6021, Jun. 2019.
- [5] H. Dorch, "Request for orbital deployment and operating authority for the SpaceX Gen2 NGSO satellite system." 2022. [Online]. Available: <https://docs.fcc.gov/public/attachments/FCC-22-91A1.pdf>
- [6] S. Kozhaya and Z. M. Kassas, "A first look at the OneWeb LEO constellation: Beacons, beams, and positioning," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 5, pp. 7528–7534, Oct. 2024.
- [7] S. Li, Q. Wu, and R. Wang, "Dynamic discrete topology design and routing for satellite-terrestrial integrated networks," *IEEE/ACM Trans. Netw.*, vol. 32, no. 5, pp. 3840–3853, Oct. 2024.
- [8] F. Baccelli, S. Candel, G. Perrin, and J.-L. Puget, "Large satellite constellations: Challenges and impact," Ph.D. dissertation, Académie des Sciences, Paris, France, 2024.
- [9] J. Li, K. Xue, J. Liu, Y. Zhang, and Y. Fang, "An ICN/SDN-based network architecture and efficient content retrieval for future satellite-terrestrial integrated networks," *IEEE Netw.*, vol. 34, no. 1, pp. 188–195, Jan./Feb. 2020.
- [10] J. Li, K. Xue, D. S. Wei, J. Liu, and Y. Zhang, "Energy efficiency and traffic offloading optimization in integrated satellite/terrestrial radio access networks," *IEEE Trans. Wireless Commun.*, vol. 19, no. 4, pp. 2367–2381, Apr. 2020.
- [11] X. Zhang et al., "Energy-efficient computation peer offloading in satellite edge computing networks," *IEEE Trans. Mobile Comput.*, vol. 23, no. 4, pp. 3077–3091, Apr. 2024.
- [12] G. Chen, S. Qi, F. Shen, Q. Zeng, and Y.-D. Zhang, "Information-aware driven dynamic LEO-RAN slicing algorithm joint with communication, computing and caching," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 5, pp. 1044–1062, May 2024.
- [13] D. Jiang et al., "QoE-aware efficient content distribution scheme for satellite-terrestrial networks," *IEEE Trans. Mobile Comput.*, vol. 22, no. 1, pp. 443–458, Jan. 2023.
- [14] H. Wu, J. Li, H. Lu, and P. Hong, "A two-layer caching model for content delivery services in satellite-terrestrial networks," in *Proc. IEEE Glob. Commun. Conf.*, 2016, pp. 1–6.
- [15] Q. Wei, Y. Chen, Z. Jia, W. Bai, T. Pei, and Q. Wu, "Energy-efficient caching and user selection for resource-limited SAGINs in emergency communications," *IEEE Trans. Commun.*, vol. 73, no. 6, pp. 4121–4136, Jun. 2025.
- [16] D. Han, W. Liao, H. Peng, H. Wu, W. Wu, and X. Shen, "Joint cache placement and cooperative multicast beamforming in integrated satellite-terrestrial networks," *IEEE Trans. Veh. Technol.*, vol. 71, no. 3, pp. 3131–3143, Mar. 2022.
- [17] L. He, Z. Jia, K. Guo, H. Gan, Z. Han, and C. Yuen, "Online joint data offloading and power control for space-air-ground integrated networks," *IEEE Trans. Wireless Commun.*, vol. 23, no. 12, pp. 18126–18141, Dec. 2024.
- [18] H. Zhang, J. Xu, X. Liu, K. Long, and V. C. Leung, "Joint optimization of caching placement and power allocation in virtualized satellite-terrestrial network," *IEEE Trans. Wireless Commun.*, vol. 22, no. 11, pp. 7932–7943, Nov. 2023.
- [19] S. S. Hassan et al., "Satellite-based ITS data offloading & computation in 6G networks: A cooperative multi-agent proximal policy optimization DRL with attention approach," *IEEE Trans. Mobile Comput.*, vol. 23, no. 5, pp. 4956–4974, May 2024.
- [20] Z. Yang, J. Miao, T. Zhang, X. Tang, J. Kang, and D. Niyato, "QoE maximization for video streaming in cache-enabled satellite-UAV-terrestrial network," in *Proc. IEEE Int. Conf. Commun.*, 2024, pp. 1–6.
- [21] B. Mao, Y. Liu, H. Guo, J. Liu, and N. Kato, "On a hierarchical content caching and asynchronous updating scheme for non-terrestrial network-assisted connected automated vehicles," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 1, pp. 64–74, Jan. 2025.
- [22] Y. Zhao et al., "Joint content caching, service placement and task offloading in UAV-enabled mobile edge computing networks," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 1, pp. 51–63, Jan. 2025.
- [23] U. Niesen, D. Shah, and G. W. Wornell, "Caching in wireless networks," *IEEE Trans. Inf. Theory*, vol. 58, no. 10, pp. 6524–6540, Oct. 2012.
- [24] J. Tang et al., "Opportunistic content-aware routing in satellite-terrestrial integrated networks," *IEEE Trans. Mobile Comput.*, vol. 23, no. 11, pp. 10460–10474, Nov. 2024.
- [25] X. Cao et al., "Edge-assisted multi-layer offloading optimization of LEO satellite-terrestrial integrated networks," *IEEE J. Sel. Areas Commun.*, vol. 41, no. 2, pp. 381–398, Feb. 2023.
- [26] Y. Chen, J. Zhao, Y. Wu, J. Huang, and X. Shen, "Multi-user task offloading in UAV-assisted LEO satellite edge computing: A game-theoretic approach," *IEEE Trans. Mobile Comput.*, vol. 24, no. 1, pp. 363–378, Jan. 2025.
- [27] Z. Jia et al., "NFV-enabled service recovery in space-air-ground integrated networks: A matching game based approach," *IEEE Trans. Netw. Sci. Eng.*, vol. 12, no. 3, pp. 1732–1744, May/Jun. 2025.
- [28] Z. Zhang, Y. Wu, Z. Ma, X. Lei, L. Lei, and Z. Wei, "Coordinated multi-satellite transmission for OTFS-based 6G LEO satellite communication systems," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 1, pp. 156–170, Jan. 2025.
- [29] L. Breslau, P. Cao, L. Fan, G. Phillips, and S. Shenker, "Web caching and Zipf-like distributions: Evidence and implications," in *Proc. 1999 IEEE Int. Conf. Comput. Commun.*, 1999, vol. 1, pp. 126–134.
- [30] J. Li, K. Xue, J. Liu, and Y. Zhang, "A user-centric handover scheme for ultra-dense LEO satellite networks," *IEEE Wireless Commun. Lett.*, vol. 9, no. 11, pp. 1904–1908, Nov. 2020.
- [31] J. Jiao et al., "Age-optimal downlink NOMA resource allocation for satellite-based IoT network," *IEEE Trans. Veh. Technol.*, vol. 72, no. 9, pp. 11575–11589, Sep. 2023.
- [32] A. Abdi, W. C. Lau, M.-S. Alouini, and M. Kaveh, "A new simple model for land mobile satellite channels: First-and second-order statistics," *IEEE Trans. Wireless Commun.*, vol. 2, no. 3, pp. 519–528, May 2003.
- [33] R. Wünsche, M. Krondorf, and A. Knopp, "Investigations of channel capacity loss in LEO satellite systems using phased array beamforming antennas," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 4, pp. 5373–5394, Aug. 2024.
- [34] N. Heydarishahreza, T. Han, and N. Ansari, "Spectrum sharing and interference management for 6G LEO satellite-terrestrial network integration," *IEEE Commun. Surveys Tuts.*, vol. 27, no. 5, pp. 2794–2825, Oct. 2025.

- [35] X. Ding, Y. Lei, Y. Zou, G. Zhang, and L. Hanzo, "Interference management by harnessing multi-domain resources in spectrum-sharing aided satellite-ground integrated networks," *IEEE Trans. Veh. Technol.*, vol. 73, no. 6, pp. 8306–8321, Jun. 2024.
- [36] D.-H. Tran, S. Chatzinotas, and B. Ottersten, "Throughput maximization for backscatter-and cache-assisted wireless powered UAV technology," *IEEE Trans. Veh. Technol.*, vol. 71, no. 5, pp. 5187–5202, May 2022.
- [37] D. P. Bertsekas, "Nonlinear programming," *J. Oper. Res. Soc.*, vol. 48, no. 3, pp. 334–334, 1997.
- [38] X. Cai, C. Song, S. Wright, and J. Diakonikolas, "Cyclic block coordinate descent with variance reduction for composite nonconvex optimization," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 3469–3494.
- [39] P. Tseng, "Convergence of a block coordinate descent method for non-differentiable minimization," *J. Optim. Theory Appl.*, vol. 109, no. 3, pp. 475–494, 2001.
- [40] C. M. de Galland et al., "Random coordinate descent for resource allocation in open multiagent systems," *IEEE Trans. Autom. Control*, vol. 69, no. 11, pp. 7600–7613, Nov. 2024.
- [41] Z. Lai et al., "LeoCC: Making internet congestion control robust to LEO satellite dynamics," in *Proc. Int. Conf. Appl., Technol., Architectures, Protoc. Comput. Commun.*, 2025, pp. 129–146.
- [42] W. X. Zheng et al., "StarCDN: Moving content delivery networks to space," in *Proc. 2025 Int. Conf. Appl., Technol., Architectures, Protoc. Comput. Commun.*, 2025, pp. 948–962.
- [43] P. Zhang, Z. Xian, M. Liao, H. Huang, and J. Yang, "Adaptive routing mechanism for LEO satellite network based on control domain partition," *IEEE Trans. Green Commun. Netw.*, vol. 9, no. 1, pp. 70–82, Mar. 2025.
- [44] L. Chen et al., "Delay-optimal cooperation transmission in remote sensing satellite networks," *IEEE Trans. Mobile Comput.*, vol. 22, no. 9, pp. 5109–5123, Sep. 2023.
- [45] D. Zhao, X. Zhang, and M. Lee, "SatPipe: Deterministic TCP adaptation for highly dynamic LEO satellite networks," in *Proc. IEEE Int. Conf. Comput. Commun.*, 2025, pp. 1–10.
- [46] K. Poularakis et al., "Distributed caching algorithms in the realm of layered video streaming," *IEEE Trans. Mobile Comput.*, vol. 18, no. 4, pp. 757–770, Apr. 2019.
- [47] S. Altunc et al., "X-band CubeSat communication system demonstration," in *Proc. AIAA/USU Conf. Small Satellites*, no. GSFC-E-DAA-TN24039, 2015.
- [48] C. Liu et al., "A Ka-band single-chip SiGe BiCMOS phased-array transmit/receive front-end," *IEEE Trans. Microw. Theory Techn.*, vol. 64, no. 11, pp. 3667–3677, Nov. 2016.
- [49] C.-Q. Dai, S. Li, J. Wu, and Q. Chen, "Distributed user association with grouping in satellite–terrestrial integrated networks," *IEEE Internet Things J.*, vol. 9, no. 12, pp. 10244–10256, Jun. 2022.
- [50] Q. Zhang, H. Wang, Z. Feng, and Z. Han, "Many-to-many matching-theory-based dynamic bandwidth allocation for UAVs," *IEEE Internet Things J.*, vol. 8, no. 12, pp. 9995–10009, Jun. 2021.
- [51] H. Zhang, Y. Liao, and L. Song, "D2D-U: Device-to-device communications in unlicensed bands for 5G system," *IEEE Trans. Wireless Commun.*, vol. 16, no. 6, pp. 3507–3519, Jun. 2017.
- [52] A. E. Roth and M. Sotomayor, "Two-sided matching," in *Handbook of Game Theory With Economic Applications*, vol. 1. Amsterdam, The Netherlands: Elsevier, 1992, pp. 485–541.
- [53] E. Bodine-Baron et al., "Peer effects and stability in matching markets," in *Proc. Int. Symp. Algorithmic Game Theory*, 2011, pp. 117–129.
- [54] F. Pantisano, M. Bennis, S. Valentin, and M. Debbah, "Matching with externalities for context-aware user-cell association in small cell networks," in *Proc. IEEE Glob. Commun. Conf.*, 2013, pp. 4483–4488.
- [55] I. Del Portillo, B. G. Cameron, and E. F. Crawley, "A technical comparison of three low Earth orbit satellite constellation systems to provide global broadband," *Acta Astronautica*, vol. 159, pp. 123–135, 2019.
- [56] L. Qu, G. Xu, Z. Zeng, N. Zhang, and Q. Zhang, "UAV-assisted RF/FSO relay system for space-air-ground integrated network: A performance analysis," *IEEE Trans. Wireless Commun.*, vol. 21, no. 8, pp. 6211–6225, Aug. 2022.
- [57] G.-Y. Chang, C.-K. Hung, and C.-H. Chen, "A CSI prediction scheme for satellite-terrestrial networks," *IEEE Internet Things J.*, vol. 10, no. 9, pp. 7774–7785, May 2023.
- [58] "P.618: Propagation data and prediction methods required for the design of Earth-space telecommunication systems," 2023. [Online]. Available: <https://www.itu.int/rec/R-REC-P.618/>
- [59] SpaceX, "SpaceX gen2 NGSO satellite system," Federal Communications Commission, Order and Authorization 22–91, 2022. [Online]. Available: <https://docs.fcc.gov/public/attachments/FCC-22-91A1.pdf>



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